

12

NUMBER AND ALGEBRA, MEASUREMENT AND SPACE

RATIOS, RATES AND TIME

Ancient Greek mathematicians and artists discovered a 'perfect' rectangle whose dimensions were most naturally pleasing to the eye. Try drawing an ideal rectangle yourself, one that is neither too thin nor too tall. Such a rectangle has a length that is $\frac{\sqrt{5}-1}{2}$ times its width. This value, which is approximately 1.618, is called the **golden ratio** and a rectangle with these dimensions is called a **golden rectangle**. Is your rectangle's length 1.6 times its width?

The ancient Greeks used the golden ratio in art and architecture. The Parthenon in Athens was built to the dimensions of the golden rectangle. Even today, the golden rectangle is often used in graphic design and publishing, such as on website panels and sidebars.



Chapter outline

	Working mathematically				
12.01 Ratios	U	F		R	C
12.02 Simplifying ratios	U	F		R	C
12.03 Ratio problems	U	F	PS	R	
12.04 Rates	U	F		R	C
12.05 Best buys	U	F		R	C
12.06 Rate problems	U	F	PS	R	C
12.07 Travel graphs	U	F	PS	R	C
12.08 Time calculations	U	F			
12.09 24-hour time	U	F			C
12.10 Time differences	U	F	PS	R	C
12.11 Timetables	U	F	PS	R	C
12.12 World time zones*	U	F	PS	R	C

*EXTENSION

Wordbank

24-hour time Time of day written using 4 digits (instead of a.m. and p.m.), for example, 18:20

best buy When comparing different brands or sizes during shopping, this is the item with the lowest unit price and is the best value for money

Quiz
Wordbank 12

rate A measurement comparing quantities of different types, expressed using 2 units

ratio A relationship between 2 or more quantities measured in the same units; for example, the ratio of 3 teachers to 40 students is 3 : 40

speed A rate that compares distance travelled with time taken

unit price The price of one item or unit, such as 1 mL or 1 g

unitary method A method of finding a quantity y finding one part firs

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (13):

12.01 Why do we have ratios?
12.02 Equivalent ratios • Simplifying ratios 1 • Simplifying ratios 2
12.03 Ratio problems 1 • Ratio problems 2
12.05 Unit pricing
12.06 Rate problems 1 • Rate problems 2
12.07 Travel graphs
12.08 Time
12.09 24-hour time
12.10 Time differences

Twig videos (3):

12.02 Ratios: The maths of baking
12.03 Ratios: Currency exchange
12.10 Time zones

PhET interactives (3):

12.01 Ratio and proportion
12.03 Proportion playground
12.04 Unit rates

Quizzes (6):

- Wordbank 12
- SkillCheck 12
- Mental skills 12A
- Mental skills 12B
- Language of maths 12
- Test yourself 12

Skillsheets (3):

12.07 Units of time • Telling the time • 24-hour time

Worksheets (19):

12.02 Ratios match • Simplifying ratios
12.03 Ratio calculations
12.04 Rates
12.06 What's my speed? • Speed • Rate problems • Ratios and rates review
12.07 Travel graph stories • The hare and the tortoise
12.08 Converting time units • What's my speed?
12.09 TV times • 12- and 24-hour time • 24-hour time on an analogue clock
12.10 Time • Time calculations • World time zones
12.11 Tide chart
12.12 World time zones
Mind map Ratios, rates and time

Puzzles (9):

12.02 Simplifying ratios
12.05 Best buys puzzle
12.06 Rate problems
12.07 Jane's diary
12.08 Converting time units
12.09 12- and 24-hour time
12.10 Time calculations • Australian times
12.11 Australian times
Language of maths Ratios and rates find-a-word

Presentation (1):

12.10 Time calculations

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- compare quantities using ratios and rates
- simplify ratios and solve problems involving ratios
- simplify rates and solve problems involving rates
- calculate unit prices to determine 'best buys'
- interpret and draw travel graphs, including examining speed
- add and subtract time mentally and using a calculator
- convert between 12-hour time and 24-hour time
- solve problems involving time differences
- interpret and use timetables
- (EXTENSION) interpret and use international time zones

SkillCheck ANSWERS ON P. 604

1 Copy and complete:

a 2000 kg = _____t **b** 360 cm = _____m **c** 18 g = _____mg

d 300 s = _____min **e** 6.5 cm = _____mm **f** 480 min = _____h

g 7500 mL = _____L **h** 8.15 km = _____m **i** 850 mm = _____cm

2 Find the highest common factor (HCF) of each pair of numbers.

a 12 and 18 **b** 8 and 36 **c** 20 and 64

3 Complete each pair of equivalent fractions.

a $\frac{7}{10} = \frac{\square}{30}$ **b** $\frac{3}{5} = \frac{12}{\square}$ **c** $\frac{18}{45} = \frac{\square}{5}$

4 Simplify each fraction.

a $\frac{6}{24}$ **b** $\frac{40}{50}$ **c** $\frac{9}{36}$

Quiz
SkillCheck 12

Ratios 12.01

A **ratio** consists of 2 or more numbers that compare the parts or shares of things of the same type, in the same units. For example, if a cake recipe uses sugar and flour in a ratio of 1 to 2, written '1 : 2,' it means that for every one part of sugar we need 2 parts of flour.

Each number in a ratio is called a **term** of the ratio.



Shutterstock.com/Impact Photography

Video
Why do we have ratios?

Interactive
Ratio and proportion

Ratios

A **ratio** compares quantities of the same type given in the same units.

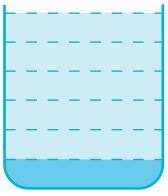
We say 'a to b' and write $a : b$, where a and b are numbers.

Jackson mixes cordial and water in the ratio 1 : 5, meaning that for every 1 part of cordial there are 5 parts of water.

Saba prefers a stronger flavour, so she uses a cordial to water ratio of 1 : 3.

For a picnic, Saba can make a bottle of her drink by mixing one **cup** of cordial to 3 **cups** of water.

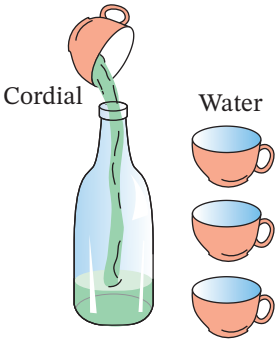
For a party, Saba can make a tub of her drink by mixing 1 **bucket** of cordial to 3 **buckets** of water.



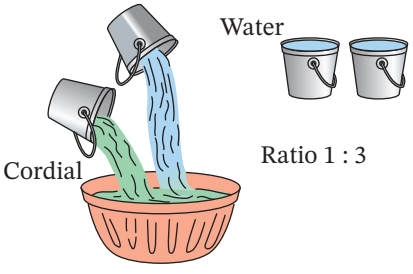
1 : 5 (read as '1 to 5')
1 part cordial to 5 parts water



1 : 3 (read as '1 to 3')
1 part cordial to 3 parts water



Ratio 1 : 3



Ratio 1 : 3

Example 1

For this box of chocolates, write each ratio.

a Dark brown chocolates to light brown chocolates

b Rectangular chocolates to circular chocolates

c White-topped chocolates to all chocolates



Shutterstock.com/Richard Semik

Solution

a Dark chocolates to light chocolates = 4 : 10 — 4 dark, 10 light

b Rectangular chocolates to circular chocolates = 6 : 2 — 6 rectangular, 2 circular

c White-topped chocolates to all chocolates = 6 : 20 — 6 white, 20 in total

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Example 2

Express each pair of quantities as a ratio.

a A mass of 69 kg to a mass of 80 kg

b A distance of 37 cm to a distance of 1 metre

Solution

a Since the quantities are in the same units (kg) the ratio is simply 69 : 80.

69 kg : 80 kg = 69 : 80

b Change 1 metre to 100 centimetres so that both quantities are in the same units.

37 cm : 1 m = 37 cm : 100 cm

= 37 : 100

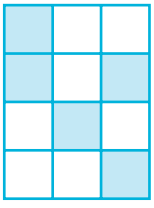
EXERCISE 12.01 ANSWERS ON P. 604

Ratios UFRC

EXAMPLE 1

1 For each diagram, write the ratio of shaded to unshaded parts. C

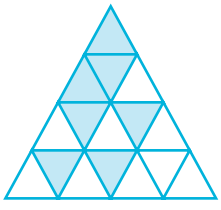
a



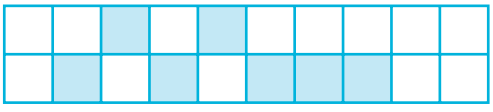
b



c



d

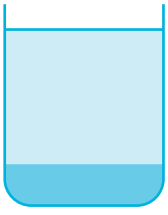


e



2 **a** Joe likes to make a drink with a ratio of 2 parts cordial to 5 parts water (or 2 : 5). Draw a diagram to show this ratio in a glass.

b Melanie made a drink mixed with the ratio shown in the diagram. Estimate the ratio of cordial to water used.



3 In your classroom, find the ratio of: C

a teachers to students

b girls to boys

c chairs to tables

d students to chairs

4 In a class of 25 students, Hadid found that 13 students in his class have brown eyes, 8 students have blue eyes, 3 students have green eyes and 1 student has hazel eyes. Find the ratios between the student eye colours described. C

a blue to brown

b green to blue

c brown to hazel

d hazel to the total

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5 Out of every 100 people at the football, 39 were men, 23 were women, and the rest were children. Find each ratio. C

a men to women
b women to the total
c children to men
d women to children

EXAMPLE 2

6 Express each pair of quantities as a ratio. C

a 13 g to 25 g
b $17 to $100
c 5 litres to 12 litres
d 150 km to 1 km
e 0.5 kg to 43 g
f $3.50 to 83 cents
g 3 kL to 1421 L
h 27 mm to 4 cm
i 59 minutes to 3 hours
j 15 years to 7 months
k 9 days to 7 weeks
l 77 m to 17 cm

7 What is 39 kg to 41 g expressed as a ratio? Select the correct answer **A**, **B**, **C** or **D**. C

A 39 : 41
B 41 : 39
C 39 : 41 000
D 39 000 : 41

8 Write each statement as a ratio. C

a For each teacher there were 25 students.
b To cook steamed rice, you need 1 cup of rice to 2 cups of water.
c Adventure cycling activities in schools require one teacher for every 4 students.
d There are 11 girls for every 9 boys in the school.
e No matter how much they won on Lotto, Khloe and Kim agreed to share the prize equally.
f She was twice as tall as he was.

9 In the 1950s, the average ratio of teachers to students in classrooms was 1 : 45, in the 1970s it was 1 : 35, and in the 1980s the ratio was set at 1 : 30 for most classes. Today, it is 1 : 24. What does it mean when we say that the teacher to student ratio has improved? R C

Did you know?

Ratios in many careers

Whatever career you follow, the chances are you will need to use ratios.

- A hairdresser mixes hair colour using a ratio of 1 part colour and 1.5 parts developer
- A chef uses ratios to mix the ingredients for a sauce
- In sports analysis, win/loss ratios or improvement ratios are calculated to judge performance
- A plumber uses ratios to calculate water flow and pipe size
- A financial analyst uses ratios involving investment and profit/loss to evaluate a company's performance
- Builders use ratios when mixing mortar or concrete
- Paint colours are mixed with white paint and different ratios of colours

Find out what ratio of metals is used to make Australian coins.

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Simplifying ratios

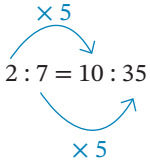
12.02

Equivalent ratios are *equal* ratios and can be found in a similar way to that used for finding **equivalent fractions**.

Equivalent ratios

To find an equivalent ratio, multiply or divide each term by the same number.

For example, 2 : 7 and 10 : 35 are equivalent ratios.



Example 3

Complete each pair of equivalent ratios.

a 6 : 4 = 18 : _____ **b** 3 : 4 = _____ : 24 **c** 12 : 8 = 3 : _____

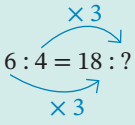
Solution

a To find the missing term, look at the 2 known matching terms, 6 and 18.

6 is *multiplied* by 3 to give 18, so do the same thing to the 4.

$4 \times 3 = 12$

So 6 : 4 = 18 : 12

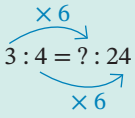


b The 2 known matching terms are 4 and 24.

4 is *multiplied* by 6 to give 24, so do the same thing to the 3.

$3 \times 6 = 18$

So 3 : 4 = 18 : 24

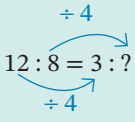


c The 2 known matching terms are 12 and 3.

12 is *divided* by 4 to give 3, so do the same thing to the 8.

$8 \div 4 = 2$

So 12 : 8 = 3 : 2



Worksheets
Ratios match
Simplifying ratios

Puzzle
Simplifying ratios

Videos
Equivalent ratios
Ratios: The maths of baking

Simplifying ratios

- To **simplify** a ratio, keep dividing both terms by the same number, preferably a large number such as their highest common factor (HCF), until each term is as small as possible
- If all terms are even, divide by 2 or perhaps 4
- Otherwise, try dividing by the odd numbers 3, 5 or 7

Videos
Simplifying ratios 1
Simplifying ratios 2

Example 4

Simplify each ratio.

a 5 : 10 **b** 27 : 18 **c** 12 : 20 : 16

Solution

a $5 : 10 = \frac{5}{5} : \frac{10}{5}$ Divide both terms by their HCF, 5

$= 1 : 2$

Or enter 5 : 10 as a fraction $\frac{5}{10}$ on the calculator: 5 $\square\!\square$ 10 =

On some calculators, use the $a^{b/c}$ key instead of $\square\!\square$

b $27 : 18 = \frac{27}{9} : \frac{18}{9}$ Divide both terms by their HCF, 9

$= 3 : 2$

Or enter 27 : 18 as an improper fraction $\frac{27}{18}$ on the calculator:
27 $\square\!\square$ 18 =

Then to change the mixed numeral to an improper fraction, press:
SHIFT S⇔D or 2nd F $a^{b/c}$

c $12 : 20 : 16 = \frac{12}{4} : \frac{20}{4} : \frac{16}{4}$ Divide all terms by their HCF, 4

$= 3 : 5 : 4$

Because there are more than 2 terms in this ratio, the calculator cannot be used here

Example 5

Simplify 10 hours : 1 day.

Solution

10 hours : 1 day = 10 hours : 24 hours Change to the same units

$= 10 : 24$

$= \frac{10}{2} : \frac{24}{2}$

$= 5 : 12$

EXERCISE 12.02 ANSWERS ON P. 605

Simplifying ratios U F R C

1 Copy and complete each pair of equivalent ratios.

EXAMPLE 3

a	2 : 3 = 8 : _____	**b**	1 : 5 = 2 : _____	**c**	3 : 5 = _____ : 15
d	4 : 7 = _____ : 35	**e**	5 : 8 = 20 : _____	**f**	7 : 12 = 49 : _____
g	5 : 11 = _____ : 66	**h**	3 : 4 = _____ : 100	**i**	2 : 1 = 10 : _____
j	_____ : 9 = 20 : 36	**k**	12 : _____ = 3 : 1	**l**	_____ : 45 = 6 : 9
m	24 : 12 = 4 : _____	**n**	16 : _____ = 2 : 5	**o**	_____ : 20 = 15 : 60

12.02

2 Which ratio is not equivalent to 32 : 48? Select the correct answer **A**, **B**, **C** or **D**.

A	16 : 24	**B**	4 : 6	**C**	2 : 3	**D**	6 : 8

3 Simplify each ratio.

EXAMPLE 4

a	10 : 100	**b**	8 : 16	**c**	12 : 30	**d**	35 : 49
e	18 : 12	**f**	56 : 24	**g**	1000 : 100	**h**	45 : 99
i	7 : 7	**j**	20 : 48	**k**	56 : 14	**l**	3 : 36
m	8 : 12 : 20	**n**	15 : 20 : 30	**o**	27 : 9 : 36	**p**	9 : 6 : 24 : 12

4 Simplify 8 months : 2 years. Select **A**, **B**, **C** or **D**.

A	8 : 2	**B**	4 : 1	**C**	1 : 3	**D**	2 : 5

5 Simplify each ratio.

EXAMPLE 5

a	80 cm : 2 m	**b**	300 g : 1.2 kg	**c**	6 days : 4 weeks
d	45 min : 2 hours	**e**	70c : $2.10	**f**	2 years : 15 months
g	15 hours : 2 days	**h**	20 mm : 1 m	**i**	4 t : 350 kg
j	250 mL : 3 L	**k**	1800 m : 1 km	**l**	8 kg : 400 g
m	2 min : 8 s	**n**	750 L : 5 kL	**o**	$18 : $4.50

6 Simplify 4 m to 25 cm. Select **A**, **B**, **C** or **D**.

A	4 : 25	**B**	400 : 25	**C**	8 : 5	**D**	16 : 1

7 The following amounts of books were sold at a garage sale: 48 crime novels, 20 science fiction novels, 35 non-fiction books, 28 romance novels.

Express each ratio in simplest form: C

a crime to romance

b sci-fi to non-fiction

c non-fiction to romance

d sci-fi to crime

e romance to non-fiction

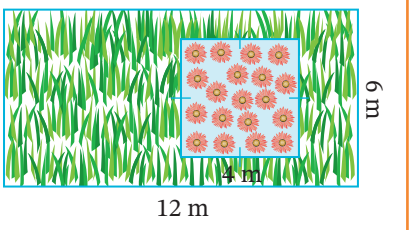
f crime to all books



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8 A backyard contains a square garden, as shown. What is the ratio of garden area to backyard area? Select **A**, **B**, **C** or **D**. C



A 2 : 7 **B** 2 : 9

C 7 : 2 **D** 9 : 2

9 At Westvale Catholic College there are 72 teachers and 1440 students. Of these, 624 students are girls. R C

a Find each ratio.

i teachers : students **ii** boys : girls **iii** boys : teachers

b The government states that the teacher to student ratio must be 1 : 25 or higher. Does the college have enough teachers? Give reasons for your answer.

10 When comparing the rate at which babies are born in different countries (that have different population sizes), we use a base number of 1000. The annual birth rate in Australia is approximately 12 per 1000 people (a ratio of 12 : 1000). The rate in Pakistan is approximately 30 per 1000 people, or 30 : 1000. Express each ratio in simplest form. C

11 Tom and Ahmed bought a scooter together, but Tom paid $70 and Ahmed paid $55. Write the ratio of their shares of the scooter in simplest form. C

12 To make a 2 L jug of cordial drink, 375 mL of cordial is mixed with water. Find: C

a the amount of water used

b the ratio of cordial to water

c the ratio of cordial to drink



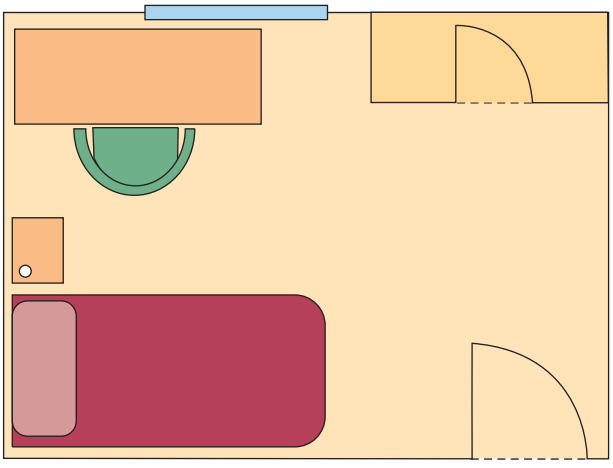
Alamy Stock Photos/Tim Allen

13 3 Year 7 classes were surveyed to find the ratio of left-handed to right-handed students. In 7N, the ratio was 1 : 5, in 7P the ratio was 2 : 3, and in 7D the ratio was 1 : 6. Which class had the highest proportion of left-handed students? Justify your answer. R C

Investigation

Scale diagrams

Here is a scale plan of a bedroom. The scale ratio is 1 : 50, meaning that 1 cm on the plan represents 50 cm on the actual bedroom. The actual bedroom is 50 times larger.



1 By measuring the plan, find in metres:
   - **a** the width of the bedroom
   - **b** the length of the bedroom

2 Calculate:
   - **a** the perimeter of the bedroom in metres
   - **b** the area of the bedroom in square metres

3 By measuring the plan, find in centimetres:
   - **a** the length of the bed
   - **b** the width of the desk
   - **c** the width of the door
   - **d** the length of the window

Ratio problems 12.03

Problems involving ratios can be solved using **equivalent ratios** or the **unitary method**. With the unitary method, we find the size of **one part** first.

Example 6

The ratio of boys to girls in a Year 7 class is 2 : 3. If there are 10 boys in the class, how many girls are there?

Solution

Method 1: Equivalent ratios

Write the problem as a pair of equivalent ratios.

Videos
Ratio problems 1
Ratio problems 2
Currency exchange

Interactive
Proportion playground

Worksheet
Ratio calculations

Boys : girls = 2 : 3 = 10 : ____ (× 5 above, × 5 below)

2 is multiplied by 5 to give 10, so do the same thing to 3.

Number of girls = 3 × 5 = 15

There are 15 girls in the class.

Method 2: Unitary method

Boys : girls = 2 : 3

2 parts (boys) = 10

1 part = 10 ÷ 2 = 5

'Unitary' means 'one'

3 parts (girls) = 3 × 5 = 15

There are 15 girls in the class.

Finding one part first

Example 7

To make heavy duty concrete, a builder mixes sand and cement in the ratio 5 : 4.
If a mix of concrete contains 24 kg of cement, find:

a the amount of sand in the mix

b the total mass of the mix.

Solution

a **Method 1: Equivalent ratios**

4 is multiplied by 6 to give 24, so do the same thing to 5.

sand : cement = 5 : 4 = ____ : 24 (× 6 above, × 6 below)

Amount of sand = 5 × 6 = 30 kg

Method 2: Unitary method

Sand : cement = 5 : 4

4 parts (cement) = 24 kg

1 part = 24 ÷ 4 = 6 kg

5 parts (sand) = 5 × 6 = 30 kg

b Total mass = 24 kg + 30 kg = 54 kg

So the total mixture was 54 kg.

Cement and sand

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EXERCISE 12.03 ANSWERS ON P. 605

Ratio problems U F PS R

EXAMPLE 6

1 In Year 7, the ratio of boys to girls is 4 : 5. If there are 64 boys in Year 7, find the number of girls.

2 Simon and Joshua's heights are in the ratio 8 : 9. If Joshua is 171 cm tall, how tall is Simon?

3 In a rectangle, the ratio of the width to the length is 5 : 12. If the length of the rectangle is 48 cm, find the width.

4 A car dealer finds that the ratio of sedans to hatchbacks sold last week was 15 : 4. If 105 sedans were sold, how many hatchbacks were sold?

EXAMPLE 7

5 A tiler uses 4 green tiles to every 3 white ones. How many tiles are used altogether if 140 green tiles are used? PS

6 When making concrete for the foundations of buildings, sand and cement are mixed together in the ratio 4 : 1. If 120 kg of cement has been delivered, what mass of sand is needed? PS



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7 Alison and Elena buy a length of material and divide it between them in the ratio 2 : 3. If Alison has 3.6 m, what length of material does Elena have?
Select the correct answer **A**, **B**, **C** or **D**. PS

A 1.8 m **B** 2.4 m **C** 5.4 m **D** 9 m

8 An alloy contains copper and iron in the ratio 2 : 5. A quantity of alloy contains 20 kg of iron. What mass of copper does it contain? PS

9 The ratio of adults to children at the cinema was 3 : 4. If there were 96 children, how many adults are there? PS

10 Enzo's Produce Store buys fruit and vegetables in the ratio 5 : 7. The mass of fruit is 8.5 tonnes. What is the total mass of produce ordered? PS



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11 Mai, Bree and Felicity share a Lotto prize in the ratio 10 : 8 : 7. Bree received $72. PS R

a How much did Mai and Felicity each receive?

b What was the total prize money shared?

12 In a triangle, the lengths of the sides are in the ratio 3 : 4 : 5. If the longest side is 30 cm long, find the perimeter of the triangle. PS R

13 In an outback mining town, the ratio of women to men is 2 : 5. If there are 240 women, how many men and women are in the town altogether? Select **A**, **B**, **C** or **D**. PS R

A 840 **B** 600 **C** 960 **D** 1680

14 To make Superglue, the contents of Tube A and Tube B are mixed in the ratio 3 : 1. PS

a If 15 mL of Tube A is used, how much of Tube B is needed?

b How much glue is made altogether if 15 mL of Tube B is used?

15 The heights of 2 buildings are in the ratio 7 : 6. If the shorter building is 150 m tall, how high is the taller building?

16 To make lighter concrete for bricklaying, Jo mixes 1 part cement to 2 parts gravel to 4 parts sand. If she uses 50 kg of gravel, how much cement and sand does she need? PS R

17 Flour and butter are mixed in the ratio 3 : 1 to make short crust pastry. If Adam makes 800 g of pastry, how much flour does he need? PS R

18 Sonya and Xanthe share a prize of $495 in the ratio of 4 : 5. How much prize money does each person receive? PS R

Rates

12.04

While a ratio compares 2 or more quantities measured in the same units, a **rate** compares 2 quantities measured in **different units**.

A rate shows how one quantity changes with another quantity. We write a rate using a '/' symbol in the form 'something *per* something else'. For example:

- the price of petrol is stated in cents per litre, or c/L
- your heart beats at a rate measured in beats per minute, or beats/min
- the **speed** of a car is measured in kilometres per hour, or km/h.

The word 'per' means 'for each' so we express a rate 'per single unit'. For instance, travelling at a rate of 50 km/h means travelling 50 km in *each* hour.

Worksheet
Rates

Interactive
Unit rates

12.04

Example 8

Write each statement as a simplified rate.

a An Internet connection costs $88.50 for 3 months.

b A car travels 820 km in 8 hours.

c A garden bed needs 6 kg of fertiliser for 5 square metres.

Solution

a The cost rate would be expressed in dollars per month.

$$\text{Connection rate} = \frac{\$88.50}{3\,\text{months}}$$

We can write a rate as a fraction: divide the number of dollars by the number of months

$$= \$29.50/\text{month}$$

b The car's speed would be expressed in kilometres per hour.

$$\text{Speed} = \frac{820\,\text{km}}{8\,\text{hours}}$$

Divide the number of kilometres by the number of hours

$$= 102.5\ \text{km/h}$$

c The rate would be expressed in kg per m^2.

$$\text{Fertiliser cover} = \frac{6\,\text{kg}}{5\,\text{m}^2}$$

Divide the number of kilograms by the number of square metres

$$= 1.2\ \text{kg/m}^2$$

EXERCISE 12.04 ANSWERS ON P. 605

Rates U F R C

1 Write the units suitable for each rate, in the form _____/_____. R C

a typing speed
b heart rate
c cost of a mobile phone call
d cost of meat
e a worker's wage
f water filling a swimming pool
g cost of posting a parcel
h population growth
i the cost of water

EXAMPLE 8

2 Write each statement as a simplified rate: C

a 180 km in 3 hours
b $26.40 for 4 hours
c 100 m in 8 seconds
d $12 for 750 mL
e 76 goals in 4 games
f $48 for 2.5 kg
g 180 minutes for 10 questions
h 3000 words in 50 minutes
i 36 children in 20 families
j 9.8 m in 1 second
k 9000 revolutions in 60 seconds
l 36 runs in 8 overs
m $1.60 for 32 grams
n 12 L to travel 100 km
o 50 sheep in 2.5 hours
p 98 containers in 7 days

3 A lift should carry no more than 1600 kg or 20 people. What is this weight allowance, in kg/person? Select the correct answer **A**, **B**, **C** or **D**.

A 8 **B** 80 **C** 0.125 **D** 32 000

4 In your own words, explain what is meant by each rate. C

a a postage cost of $1.40/kg
b a farmer keeping 60 sheep/hectare
c a factory manufacturing 920 toys/day
d a pump supplying 230 L/hour

5 The cost of 38 litres of petrol is $71.82. Express this cost in c/L. C

6 A tree grew 27 metres in 60 years. Calculate its growth rate in cm/year. C

7 A complaints hotline took 852 calls in one year. Calculate the number of calls per month. C

8 A pulp mill clears 27 000 hectares of forest in 15 years. At what rate in hectares/year does the mill clear the forest? C

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Mental skills 12A: Maths without calculators ANSWERS ON P. 605



Quiz
Mental skills 12A

12.04

Multiplying by 5, 15, 25 and 50

It is easier to multiply a number by 10 than by 5, so to multiply by 5, we can halve the number, then multiply by 10. This is because $\frac{1}{2} \times 10 = 5$. We can use a similar strategy to multiply by 15, 25 and 50.

1 Study each example.

a To multiply by 5, halve the number, then multiply by 10.

$$\begin{aligned} 18 \times 5 &= 18 \times \frac{1}{2} \times 10 \qquad \text{(or factorise 18 into } 9 \times 2\text{)} \\ &= 9 \times 10 \\ &= 90 \end{aligned}$$

b To multiply by 50, halve the number, then multiply by 100.

$$\begin{aligned} 26 \times 50 &= 26 \times \frac{1}{2} \times 100 \qquad \text{(or factorise 26 into } 13 \times 2\text{)} \\ &= 13 \times 100 \\ &= 1300 \end{aligned}$$

c To multiply by 25, quarter the number, then multiply by 100.

$$\begin{aligned} 44 \times 25 &= 44 \times \frac{1}{4} \times 100 \qquad \text{(or factorise 44 into } 11 \times 4\text{)} \\ &= 11 \times 100 \\ &= 1100 \end{aligned}$$

d To multiply by 15, halve the number, then multiply by 30.

$$\begin{aligned} 8 \times 15 &= 8 \times \frac{1}{2} \times 30 \qquad \text{(or factorise 8 into } 4 \times 2\text{)} \\ &= 4 \times 30 \\ &= 120 \end{aligned}$$

2 Now simplify each product.

a	32×5	**b**	14×5	**c**	48×5	**d**	18×50
e	52×50	**f**	36×25	**g**	44×50	**h**	12×25
i	12×15	**j**	16×25	**k**	22×50	**l**	14×15
m	28×25	**n**	36×50	**o**	26×15	**p**	22×35

12.05 Best buys

Video Unit pricing

Puzzle Best buys puzzle

When shopping, it is important to compare the prices of different brands or sizes of items and calculate which is the **best buy** ('best value for money'). This can be done by comparing the **unit price** (cost of one item or unit) of each brand or size and choosing the cheapest one. A unit could be one gram or one millilitre.

Unit price

Unit price = cost ÷ number of items or units

Supermarkets are required by law to display unit prices to allow you to compare brands or sizes of items. If you look carefully at price tags on their shelves, you will notice unit prices displayed in small print.

Example 9

Which brand of corn chips is the best buy?

- Cruncho: 50 g for $1.02
- Hombre: 125 g for $2.20
- Mexicana: 200 g for $4.81

Solution

Calculating the unit prices (cost per gram):

Cruncho = $1.02 ÷ 50 = $0.0204/g

Hombre = $2.20 ÷ 125 = $0.0176/g

Mexicana = $4.81 ÷ 200 = $0.024/g

Hombre has the lowest unit price, so it is the best buy.

Example 10

Which size of fruit juice is the better buy?

- Small bottle: $3.20 for 600 mL
- Large carton: $8.05 for 2.25 L

Solution

Calculating the unit prices (cost per mL):

Small bottle = $3.20 ÷ 600

= $0.00533333... /mL

Large carton = $8.05 ÷ 2250 2.25 L = 2250 mL

= $0.00357777... /mL

The large carton of fruit juice has the lower unit price, so it is the better buy.

EXERCISE 12.05 ANSWERS ON P. 605

Best buys U F R C

1 Calculate the unit price of each item, correct to the nearest cent. R C

a 5 razors for $4.50
b 6 eggs for $3.19
c 8 juice packs for $4.45
d 80 teabags for $3.90
e 200 g biscuits for $2.79
f 40 g lollies for $1.20

EXAMPLE 9

2 Find the better buy in each pair of items. R C

a Block ices: 18-pack for $5.00, Frooty Licious: 8-pack for $2.40
b Large salad: 400 g for $4.80, Medium salad: 250 g for $2.80
c Crunchy chips single pack: 45 g for $2.10, party pack: 150 g for $6.50
d Moo Milk: 400 mL for $0.78, or 3 L for $5.20
e Peanut butter: 80 g for $2.20, or 280 g for $6.50
f Super USB drives: 6 for $15.50, Mega USB drives: 8 for $20

EXAMPLE 10

3 For each item, find which brand gives the best buy. R C

		Sols	Baldy	GIA	Bullworths
a	Salt	500 g for $1.89	200 g for 76 c	100 g for 39 c	1 kg for $3.74
b	Juice	250 mL for $1	50 mL for 17 c	1 L for $2	200 mL for 70 c
c	Bananas	750 g for $5.20	400 g for $2.40	250 g for $1.70	2 kg for $12.20
d	Aluminium foil	20 m for $3.29	5 m for $0.80	30 m for $4.90	18 m for $2.85

4 For each item, find which size is the best value for money. R C

a



b



Left to right: Alamy Stock Photos/Alan Keith Beastall, Shutterstock.com/abimages, Shutterstock.com/Gts

c



d



Left to right: Smart Media/Jalna, Smart Media/Jalna, Alamy Stock Photos/Russell Hart

5 Which brand of rice is the best buy? Select the correct answer **A**, **B**, **C** or **D**. R C

A Super Grain: 400 g for $1.15
B Wonder Bowl: 600 g for $1.74
C Hearty White: 1 kg for $2.95
D The Rice Stuff: 20 kg for $59.80

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6 For each item, which size gives the better value for money? R C

		Small	Large
a	Premium mince	450 g for $5.35	600 g for $6.80
b	Spring water	300 mL for $1.84	420 mL for $2.20
c	Rice crackers	350 g for $4.20	500 g for $5.90
d	Mixed nuts	450 g for $7.60	750 g for $11.90
e	Coffee	60 g for $3.30	150 g for $6.65

7 When buying food, the cheapest brand or size may *not* actually be the best one to buy. R C

a Give one example where you wouldn't want to buy the cheapest brand.

b Give one example where you wouldn't want to buy the cheapest size.

c Generally speaking, what factors other than price would you consider when buying supermarket items?

12.06 Rate problems

Videos
Rate problems 1
Rate problems 2

Puzzle
Rate problems

Worksheets
What's my speed?
Speed
Rate problems
Ratios and rates review

Problems involving rates can usually be solved by multiplying or dividing. The following strategy may help.

- Write the units of the rate x/y as a fraction: $\frac{x}{y}$
- To find the quantity in the numerator, x, **multiply** by the rate.
- To find the quantity in the denominator, y, **divide** by the rate.

Example 11

A bus travels at an average speed of 88 km/h. How far will it travel in $3\frac{1}{2}$ hours?

Solution

The units of the rate expressed as a fraction is $\frac{\text{km}}{\text{h}}$.

To find the distance travelled in **kilometres** (the numerator), multiply by the rate.

No. of hours × distance travelled per hour

$$\text{Distance travelled} = 3\frac{1}{2} \times 88$$
$$= 308 \text{ km}$$

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Example 12

A soft drink factory makes cans at a rate of 150 cans/min.

a How long (to the nearest minute) would it take to produce 1000 cans?

b How many cans are produced in the 4-hour shift before lunch?

Solution

The units of the rate expressed as a fraction is $\frac{\text{cans}}{\text{min}}$.

a To find the number of minutes (the denominator), divide by the rate.

$\text{Time taken} = 1000 \div 150$ (How many times 150 divides into 1000)

$= 6\frac{2}{3}$ minutes

≈ 7 minutes

b To find the number of **cans** (the numerator), multiply by the rate.

$\text{No. of cans} = 4 \times 60 \times 150$ (No. of mins × cans per min)

$= 360\,000$

$4 \text{ h} = 4 \times 60 \text{ min}$

12.06

EXERCISE 12.06 ANSWERS ON P. 605

Rate problems U F PS R C

EXAMPLE 11

1 Nate types 55 words per minute. How many words can he type in 20 minutes?

2 Jade earns $37.80 per hour as a publicist. How much will she earn if she works 38 hours in a week?

3 A cricket batsman scores 35 runs per innings. How many runs will he score in 12 innings?

4 An aeroplane travelled at 920 km/h. How far did it travel in 4 hours?

5 A tap drips water at a rate of 15 mL/h. How much water will be wasted: PS R

a in one day?

b in one week?



Shutterstock.com/David Koscheck

EXAMPLE 12

6 A plane can carry a total of 450 passengers per flight. How many flights will it take to carry 2700 passengers? PS R

7 Mia cycles at 12 km per hour for 4 hours. How far does she travel?
Select the correct answer **A**, **B**, **C** or **D**.

A 3 km **B** 30 km **C** 48 km **D** 480 km

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8 A farmer can graze 15 sheep per hectare. If he has 65 hectares set aside for sheep, how many sheep can he graze? PS R

9 The flywheel of an engine rotates at a rate of 1200 revolutions per minute. How many revolutions does it make in 20 seconds? Select **A**, **B**, **C** or **D**. R

A 400 **B** 60 **C** 4000 **D** 24 000

10 Suyen runs 100 metres in 12 seconds. How far will she run in one minute at the same rate? PS R

11 The temperature in Adelaide at 7 a.m. is 8°C. If the temperature rises at a rate of 3°C/h, find:

a the temperature at 10 a.m.

b the temperature at 2 p.m.

c when the temperature will be 23°C.

12 A bus travels at an average speed of 76 km/h. Find:

a how far the bus travels in 8 hours

b how long the bus will take to travel 380 km.

13 Zac has 504 cows in a 42 hectare paddock. R C

a How many cows per hectare does he have?

b How many cows could he keep in a 50 hectare paddock at the same rate of cows per hectare?

c How many hectares would he need to keep 840 cows at the same rate?

14 Petrol costs \$1.72/L. If Justin has \$50 to spend, how much petrol could he buy (rounded down to the nearest litre)?

15 Dominic needs to buy new carpet for the family home. The carpet costs \$125/m^2 and the carpet layers charge \$480 to lay it. How much will it cost to carpet the house if Dominic needs 28 square metres of carpet? PS R

16 Zdena uses fertiliser on her farm at a rate of 30 kg/ha. How many hectares can she cover if she has 145 kg of fertiliser? Answer to the nearest 0.1 ha. PS R

17 Best Mince costs \$13/kg during a sale. Find:

a the cost of buying 1.5 kg of Best Mince

b the amount of mince that could be bought for \$22.10.

18 The distance between Sydney and Perth is 3972 km by road and 3276 km by air. How long would it take to travel this distance if you: PS R

a walked at 5 km/h? **b** jogged at 9 km/h?

c cycled at 25 km/h? **d** took the Indian Pacific train at 85 km/h?

e drove at 100 km/h? **f** flew at 900 km/h?

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Technology

Conversion rates

In this activity, you will use a spreadsheet to convert between 2 different units.

Miles to kilometres

An Imperial unit of length that is still used in some countries is the **mile**. It is a unit of measure for long distances and is approximately equal to 1.6 kilometres.

1 Set up the spreadsheet shown below to convert between miles and kilometres. In cell C2, enter the formula **=C1*1.6**.

	A	B	C	D	E	F	G	H
1	Number of Miles	0	2	4	6	8	10	
2	Number of Kilometres	0	=C1*1.6					
3								

2 Use **Fill Right** from cell C2 to convert from miles to kilometres for each of the cells from D2 to G2.

3 Extend your table to calculate the equivalent number of kilometres for:

a 12 miles **b** 20 miles **c** 30 miles

4 Now use your table to find the (approximate) equivalent number of miles for:

a 16 kilometres **b** 22 kilometres **c** 40 kilometres

Australian dollars to Hong Kong dollars

Another commonly-used rate is currency conversion. We will convert between Australian dollars and Hong Kong dollars using a spreadsheet.

1 Set up the spreadsheet shown to convert between Australian dollars (AUD) and Hong Kong dollars (HKD). 1 AUD is equal to about 5.3 HKD. In cell B3, enter the formula **=C1*5.3**.

2 Use **Fill Down** from cell B3 to convert from Australian to Hong Kong dollars for each of the cells down to B7.

3 Extend your table to calculate the equivalent number of Hong Kong Dollars for:

a 300 AUD **b** 450 AUD **c** 700 AUD

N23

	A	B
1	Australian Dollars	Hong Kong Dollars
2	0	0
3	50	
4	100	
5	150	
6	200	
7	250	
8		

4 Now use your table to find the (approximate) Australian dollars for:

a 1000 HKD **b** 1500 HKD **c** 2500 HKD

5 Kiah wants to go on a holiday to Hong Kong after she finishes school. She intends to save 1500 Australian dollars from her part-time job to spend on her holiday. Extend your table to find what amount this is equal to in Hong Kong dollars.

12.07 Travel graphs

Video
Travel graphs

Puzzle
Jane's diary

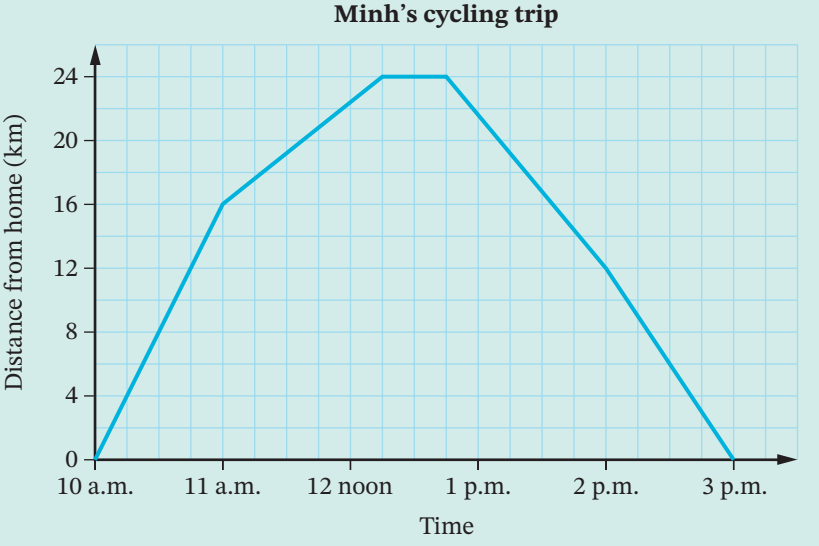
Worksheets
Travel graph stories

The hare and the tortoise

A **travel graph** or **distance–time graph** is a line graph that describes a journey, by comparing distance with time (on the vertical and horizontal axes respectively). The slope or steepness of the graph indicates speed (how fast you are travelling).

Example 13

This graph shows Minh's cycling trip.



a At what time did Minh leave home?

b When did Minh stop? How far from home was he?

c Find his speed in km/h:

i in the first hour **ii** in the second hour

d How is a change in speed shown on the graph?

e When did Minh start returning home?

f How long did it take him to return home?

Solution

a Minh left home at 10 a.m. — At the start of the graph, when distance = 0

b He stopped at 12:15 p.m., 24 km from home. — Where the graph is flat

Where the graph is flat, the distance from home does not change, which means that Minh has stopped.

c **i** In the first hour, Minh travelled from 0 km to 16 km, so his speed was 16 km/h.

ii In the second hour, Minh travelled from 16 km to 22.5 km.

Distance travelled = 22.5 km – 16 km = 6.5 km.

So his speed was 6.5 km/h.

d A change in speed is shown by a change in the steepness of the graph. The graph is steeper in the first hour than in the second hour because the speed is greater in the first hour.

Where the graph is steeper, the speed is greater: more distance travelled in less time

e Minh started returning home at 12:45 p.m.

Where the graph goes downward

f He reached home at 3 p.m.

Time to return home = 3 p.m. – 12:45 p.m. = $2\frac{1}{4}$ hours

Travel graphs

On a travel graph:

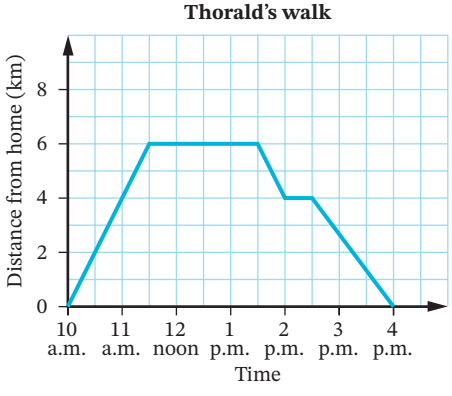
- a horizontal (flat) section on the graph indicates a stop
- the steeper the line, the greater the speed (more distance covered in less time)
- a section going down, towards the right, indicates a change in direction or that the traveller is returning towards the start.

EXERCISE 12.07 ANSWERS ON P. 606

Travel graphs U F PS R C

EXAMPLE 13

1 This travel graph shows Thorald's return trip to Emil's house. R C



a At what time did Thorald begin his trip?

b How far is Emil's house from Thorald's house?

c How long did it take Thorald to walk to Emil's house?

d Use the formula Speed = $\frac{\text{Distance travelled}}{\text{Time taken}}$ to calculate his speed in km/h during this time.

e How long did Thorald spend at Emil's house?

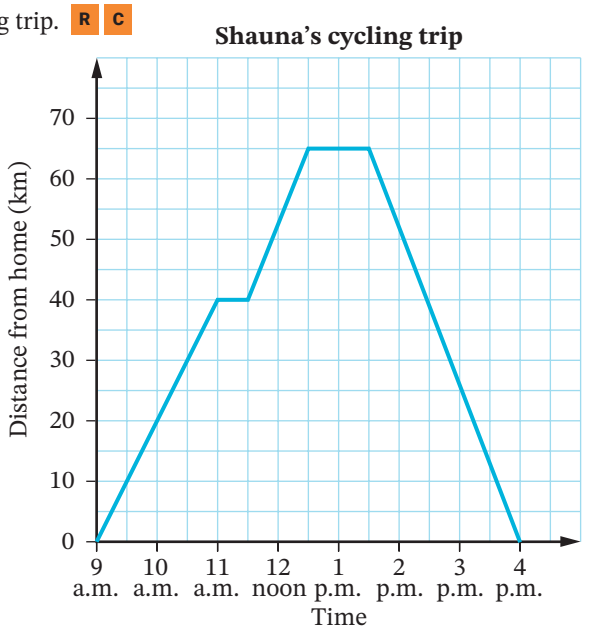
f What was the total distance that Thorald travelled?

g When and at what distance did Thorald stop on his way home?

h What was the total time of his trip, including rests?

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2 This graph shows Shauna's cycling trip. R C



a When was her first stop? How far from home was she?

b Calculate her speed in the first 2 hours.

c When did Shauna start to return home?

d How long did she take to return home?

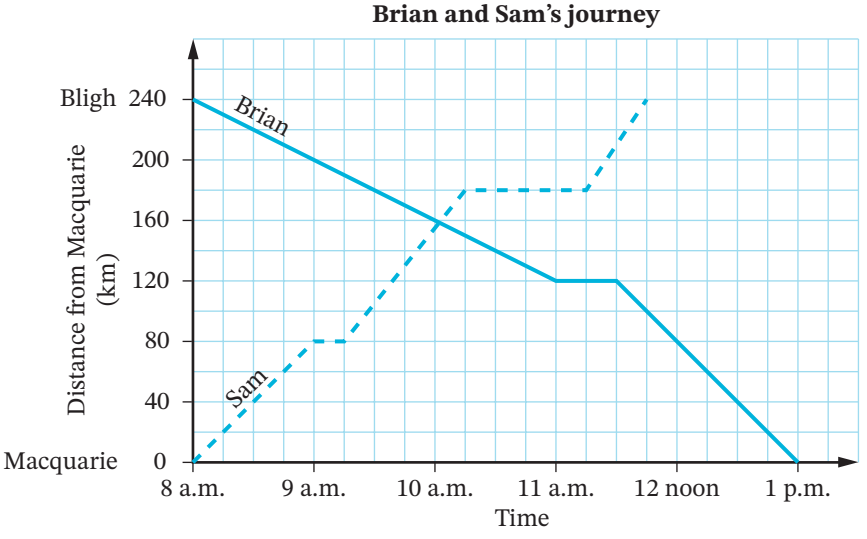
e Calculate her speed for her journey home.

f How far from home was Shauna at 3 p.m.?

g How far did Shauna travel altogether on this day?

h Did Shauna travel faster between 9 a.m. and 11 a.m., or between 11:30 a.m. and 12:30 a.m.? How does the graph illustrate this?

3 Brian travels from Bligh to Macquarie, while Sam travels from Macquarie to Bligh. R C



a How far is it between the 2 towns?

b How far is Sam from Macquarie at 10:30 a.m.?

c When does Sam reach Bligh?

d At what speed is Brian travelling before he stops?

e When do Brian and Sam pass each other? How far are they from Macquarie when they pass?

f Is Brian or Sam travelling faster during the day? How can you tell?

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g Is Brian's speed lower before 11 a.m. or after 11:30 a.m.? How can you tell?

h For how long did Sam stop altogether on the trip? Select **A**, **B**, **C** or **D**.

A 60 minutes **B** 75 minutes

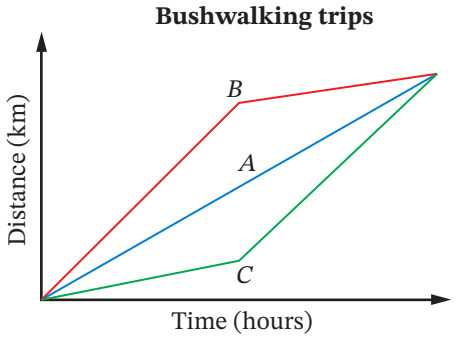
C 5 minutes **D** 90 minutes

4 Match each bushwalker described below with the correct travel graph. R C

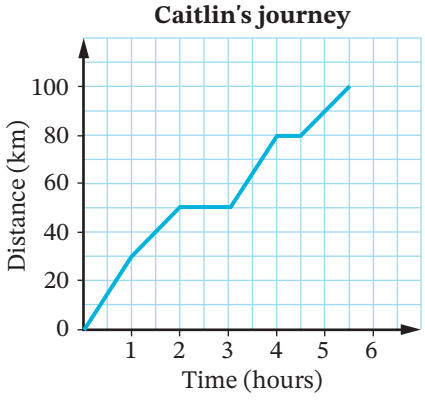
a Karlie was fast at first, but then got slower after lunch.

b Luka kept a constant speed all day.

c Tomasi was slow at first, but got faster after lunch.



5 This travel graph shows Caitlin's cycling journey. Write a story about her ride, based on the information in the graph. C



6 Jack goes on a trip to visit his grandparents. PS R C

a Draw a travel graph to represent Jack's journey:

- Jack starts at 9 a.m.
- He travels 180 km in 2 hours.
- He stops for $\frac{1}{2}$ an hour.
- He travels 120 km in $1\frac{1}{2}$ hours.
- He stops at his grandparents' house for 6 hours.
- He travels back home at 100 km/h.

b How far from Jack's home do his grandparents live?

c At what time does Jack arrive there?

d In which part of the journey is Jack travelling the fastest?

e How long does the journey home take?

f What is the total time taken for the trip?

g How far from home is Jack at 9 p.m.?

12.08 Time calculations

Worksheet
Converting time units

Puzzle
Converting time units

Video
Time

Example 14

a Round each amount of time to the nearest hour.

i 7.83 h **ii** 12 h 19 min **iii** 2 h 43 min 6 s

b Round each amount of time to the nearest minute.

i 11.4 min **ii** 25 min 37 s **iii** 3 h 6 min 30 s

Solution

a i 7.83 h ≈ 8 h

ii 12 h 19 min ≈ 12 h — Rounding down because 19 min < 30 min

When rounding to the nearest hour, use 30 minutes as the halfway mark because there are 60 minutes in an hour.

iii 2 h 43 min 6 s ≈ 3 h — Rounding up because 43 min > 30 min

b i 11.4 min ≈ 11 min

ii 25 min 37 s ≈ 26 min — Rounding up because 37 s > 30 s

When rounding to the nearest minute, use 30 seconds as the halfway mark because there are 60 seconds in a minute.

iii 3 h 6 min 30 s ≈ 3 h 7 min — Rounding up 30 s

Example 15

Complete each conversion.

a 3 years = _____ months

b 91 days = _____ weeks

c 275 min = _____ h _____ min

Solution

a 3 years = 3 × 12 months — 1 year = 12 months
= 36 months

b 91 days = (91 ÷ 7) week — 1 week = 7 days
= 13 weeks

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c $275 \text{ min} = (275 \div 60) \text{ h}$
$= 4.58333\ldots \text{ h}$
$= 4 \text{ h } 35 \text{ min}$

(or $4\frac{7}{12}$ h)

Shown on a calculator as 4°35' after entering °''' or DMS

Most scientific calculators have a degrees-minutes-seconds key (°''' or DMS) that is useful for calculations involving minutes and seconds (base 60). This key converts decimal answers for time to hours-and-minutes or minutes-and-seconds.

Example 16

Simplify each expression.

a 9 h 25 min + 6 h 48 min

b 7 h 5 min – 3 h 24 min

Solution

a $9 \text{ h } 25 \text{ min} + 6 \text{ h } 48 \text{ min} = (9+6) \text{ h} + (25+48) \text{ min}$
$= 15 \text{ h} + 73 \text{ min}$
$= 15 \text{ h} + 1 \text{ h } 13 \text{ min}$
$= 16 \text{ h } 13 \text{ min}$

OR On a calculator, enter 9 °''' 25 °''' + 6 °''' 48 °''' =

b $7 \text{ h } 5 \text{ min} - 3 \text{ h } 24 \text{ min} = (7-3) \text{ h} + (5-24) \text{ min}$
$= 4 \text{ h} + (-19) \text{ min}$
$= 3 \text{ h} + (60-19) \text{ min}$
$= 3 \text{ h } 41 \text{ min}$

OR On a calculator, enter 7 °''' 5 °''' – 3 °''' 24 °''' =

EXERCISE 12.08 ANSWERS ON P. 606

Time calculations U F

1 Round each time to the nearest hour. EXAMPLE 14

a 1 h 28 min **b** 20.7 h **c** 5 h 11 min
d 3 h 40 min 18 s **e** 9.15 h **f** 10 h 30 min

2 Round time to the nearest minute.

a 21 min 43 s **b** 7.5 min **c** 8 min 8 s
d 2 h 15 min 30 s **e** 16.31 min **f** 4 h 20 min 39 s

3 Copy and complete each conversion. EXAMPLE 15

a 5 weeks = _____ days **b** $2\frac{1}{2}$ years = _____ months
c 3 days = _____ h **d** 2 years = _____ weeks
e 1 week = _____ h **f** 3 fortnights = _____ days
g 96 h = _____ days **h** 480 s = _____ min

i 84 days = _____ weeks
j 60 months = _____ years
k 60 h = _____ days
l 56 days = _____ fortnights

4 Calculate the number of seconds in:

a one hour **b** one day **c** one week

5 Convert each time to hours and minutes.

a 340 min **b** 186 min **c** 227 min
d 405 min **e** 262 min **f** 499 min + 105 min

6 Are you over a billion seconds old? Find your age in seconds to answer this question.

7 Convert each time to minutes and seconds.

a 372 s **b** 545 s **c** 238 s
d 422 s **e** 500 s **f** 216 s + 308 s

EXAMPLE 16

8 Simplify each expression.

a 2 h 15 min + 4 h 32 min
b 9 h 20 min + 3 h 47 min
c 6 h 36 min – 2 h 11 min
d 10 h 31 min – 7 h 6 min
e 5 h 12 min + 1 h 48 min
f 3 h 45 min + 3 h 29 min
g 12 h 32 min – 8 h 19 min
h 4 h 10 min – 42 min
i 7 h 46 min + 2 h 17 min
j 5 h 33 min + 4 h 18 min
k 8 h 15 min – 6 h 24 min
l 2 h 29 min – 1 h 40 min

Worksheet
What's my speed?

Technology

Speed and distance

Jana and Sunil travel in separate cars from the same starting point. They travel at different constant speeds to the same destination. In hours, how much more quickly did Sunil arrive at their destination, compared to Jana who reached the destination in 8 hours?

1 Set up a spreadsheet as shown.

	A	B	C
1	Person:	Jana	Sunil
2	Speed (km/h):	70	80
3			
4	Time (h)	Distance travelled (km)	
5	1		
6	2		
7	3		
8	4		
9	5		
10	6		
11	7		
12	8		
13			

2 Use the formula **Distance = speed × time**. Convert this into a spreadsheet formula in cell B5 to calculate the distance that Jana has travelled after each hour has passed.

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3 Use the **Fill Down** function to copy the formula from cell B5 down to cell B12.

4 Modify the formula entered from step **2** in cell C5 to calculate the distance that Sunil travels in the first hour.

5 Use the **Fill Down** function to copy the formula from cell C5 down to cell C12.

6 By comparing the values in columns B and C, find how far Jana and Sunil's starting point was from their final destination.

7 In hours, how much more quickly did Sunil arrive at their destination, compared to Jana?

Did you know?

Minutes and seconds

In Chapter 2, you learnt that there are 360° in a revolution because the ancient Babylonians used a base 60 number system and believed that a year lasted 360 days. The Babylonians, who lived where Iraq is today, invented the units for measuring angles and time in 2000 BCE. That is why there are 60 minutes in an hour and 60 seconds in a minute.

The word 'minute' has another meaning. When pronounced '*my-newt*', it means tiny, but this meaning is also related to the minute as a unit of time. A minute is a tiny fraction of an hour, and comes from the Latin '*pars minuta prima*', meaning the first division (or part) of an hour.

Find out how the 2 meanings of the word 'second' are similarly related.

Mental skills 12B: Maths without calculators ANSWERS ON P. 606

Quiz
Mental skills 12B

Dividing by 5, 15, 25 and 50

It is easier to divide a number by 10 than by 5, so to divide by 5, we can divide by 10, then double the answer. We can use a similar strategy to divide by 15, 25 and 50.

1 Study each example.

a To divide by 5, divide by 10 and double the answer. We do this because there are 2 5s in every 10.

$140 \div 5 = 140 \div 10 \times 2$
$= 14 \times 2$
$= 28$

b To divide by 50, divide by 100 and double the answer. This is because there are 2 50s in every 100.

$400 \div 50 = 400 \div 100 \times 2$
$= 4 \times 2$
$= 8$

c To divide by 25, divide by 100 and multiply the answer by 4. This is because there are 4 25s in every 100.

$600 \div 25 = 600 \div 100 \times 4$
$= 6 \times 4$
$= 24$

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d To divide by 15, divide by 30 and double the answer. This is because there are 2 15s in every 30.

$240 \div 15 = 240 \div 30 \times 2$

$= 8 \times 2$

$= 16$

2 Now simplify each quotient.

a	$90 \div 5$	**b**	$170 \div 5$	**c**	$230 \div 5$	**d**	$330 \div 5$
e	$1300 \div 50$	**f**	$800 \div 50$	**g**	$400 \div 50$	**h**	$2200 \div 50$
i	$300 \div 25$	**j**	$1000 \div 25$	**k**	$700 \div 25$	**l**	$600 \div 15$
m	$270 \div 15$	**n**	$360 \div 45$	**o**	$900 \div 25$	**p**	$7000 \div 50$

12.09 24-hour time

Skillsheets
Units of time
Telling the time
24-hour time

Worksheets
TV times
12- and 24-hour time
24-hour time on an analogue clock

Puzzle
12- and 24-hour time

There are 2 ways to show the time of day:

- **12-hour time**: the conventional way using a.m. and p.m. and the hours 1 to 12
- **24-hour time**: a more formal way that uses the hours 0 to 23 and no a.m./p.m.

24-hour time is commonly used around the world, especially in the armed forces, navigation, in transport timetables and hospital records. This table shows the relationship between 12-hour time and 24-hour time.

24-hour time	12-hour time
00:00	12 a.m. (midnight)
01:00	1 a.m.
02:00	2 a.m.
03:00	3 a.m.
04:00	4 a.m.
05:00	5 a.m.
06:00	6 a.m.
07:00	7 a.m.
08:00	8 a.m.
09:00	9 a.m.
10:00	10 a.m.
11:00	11 a.m.

24-hour time	12-hour time
12:00	12 p.m. (midday)
13:00	1 p.m.
14:00	2 p.m.
15:00	3 p.m.
16:00	4 p.m.
17:00	5 p.m.
18:00	6 p.m.
19:00	7 p.m.
20:00	8 p.m.
21:00	9 p.m.
22:00	10 p.m.
23:00	11 p.m.

24-hour time to 12-hour time

- If it begins with '00', then it is the hour after 12:00 a.m. (midnight)
- If it begins with '12', then it is the hour after 12:00 p.m. (midday)
- If it is less than 12:00, then it is a.m. (morning time)
- If it is more than 13:00, then it is p.m. (afternoon/evening) time: subtract 12 from the hour.

12.09

Example 17

Convert each 24-hour time to 12-hour time.

a 10:20 **b** 21:30 **c** 00:50 **d** 12:40

Solution

a 10:20 < 12:00, so it is a.m. time.

10:20 = 10:20 a.m.

b 21:30 > 13:00, so it is p.m. time.

Subtract 12 from the hours: 21 – 12 = 9

21:30 = 9:30 p.m.

c 00:50 begins with 00, so it is 12:00 a.m. time.

00:50 = 12:50 a.m.

d 12:40 begins with 12, so it is 12:00 p.m. time.

12:40 = 12:40 p.m.

12-hour time to 24-hour time

- If it is the hour after 12:00 a.m., begin with '00'
- If it is a.m. time or the hour after 12:00 p.m., write as is but make sure the hour has 2 digits (for example, 02, 09)
- If it is after 1:00 p.m., then add 12 to the hour

Example 18

Convert each 12-hour time to 24-hour time.

a 8:15 a.m. **b** 2:40 p.m. **c** 12:20 p.m.

Video
24-hour time

Solution

a 8:15 a.m. is a.m. time, so write as a 4-digit number and rename 8 as 08.

8:15 a.m. = 08:15

b 2:40 p.m. is after 1:00 p.m., so add 12 to the hour: 2 + 12 = 14

2:30 p.m. = 14:30

c 12:20 p.m. is in the hour after 12:00 p.m., so write it as a 4-digit number.

12:20 p.m. = 12:20

EXERCISE 12.09 ANSWERS ON P. 606

24-hour time U F C

EXAMPLE 17

1 Convert each 24-hour time to 12-hour time. C

a	08:45	**b**	13:20	**c**	17:50	**d**	00:17
e	21:05	**f**	18:32	**g**	11:15	**h**	02:38
i	14:40	**j**	03:20	**k**	16:55	**l**	23:31
m	01:08	**n**	10:45	**o**	20:00	**p**	06:43

EXAMPLE 18

2 Convert each 12-hour time to 24-hour time. C

a	6:35 p.m.	**b**	8:05 a.m.	**c**	11:44 a.m.	**d**	11:20 p.m.
e	2:51 a.m.	**f**	12:30 p.m.	**g**	3:38 p.m.	**h**	7:11 a.m.
i	9:08 a.m.	**j**	9:54 p.m.	**k**	12:42 a.m.	**l**	7:39 p.m.
m	1:59 a.m.	**n**	10:18 p.m.	**o**	8:46 a.m.	**p**	5:23 p.m.

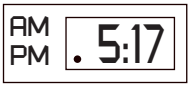
3 Write the times shown on each clock in 12-hour time. C

a 

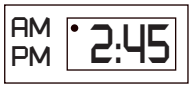
b 

c 

4 Write the times shown on each clock in 24-hour time. C

a 

b 

c 

5 This plane flight timetable shows flights from eastern Australia to Christchurch in New Zealand.

	Depart	Arrive
Sydney – Christchurch	08:55 17:10 19:45	13:55 22:10 00:45
Melbourne – Christchurch	08:45 09:05 17:55	14:05 14:25 23:15
Brisbane – Christchurch	07:50 16:30	14:15 22:55

Write the answers to each question in 12-hour time.

a When does the first flight leave Sydney and when does it arrive in Christchurch?

b When does the last flight leave Melbourne and when does it arrive in Christchurch?

c At what time between 12 midnight and 1 a.m. does a flight arrive in Christchurch?

d What is the earliest departure time?

e What is the latest departure time? Select the correct answer **A**, **B**, **C** or **D**.

A 4:30 p.m. **B** 5:10 p.m.

C 5:55 p.m. **D** 7:45 p.m.

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Time differences

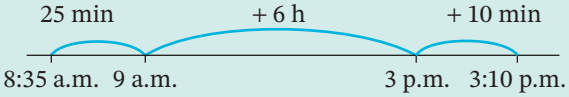
12.10

Example 19

What is the difference in time between 8:35 a.m. and 3:10 p.m.?

Solution

Method 1



From 8:35 a.m. to 9:00 a.m. = 25 minutes

From 9:00 a.m. to 3:00 p.m. = 6 hours

From 3:00 p.m. to 3:10 p.m. = 10 minutes

Total time difference = 6 h + 25 min + 10 min

= 6 h 35 min

Method 2

Convert to 24-hour time first, then use the calculator's °‘‘‘ or DMS key to enter hours and minutes, and subtract the times.

Time difference = 3:10 p.m. − 8:35 a.m.

= 15:10 – 08:35 15 °‘‘‘ 10 °‘‘‘ − 8 °‘‘‘ 35 °‘‘‘ =

= 6 h 35 min

Example 20

What is the difference in time between 10:50 and 19:30?

Solution

Method 1

From 10:50 to 11:00 = 10 min

From 11:00 to 19:00 = 8 h 19 – 11 = 8

From 19:00 to 19:30 = 30 min

Total time difference = 8 h + 10 min + 30 min

= 8 h 40 min

Method 2

Use the calculator's °‘‘‘ or DMS key to enter hours and minutes, and subtract the times.

Video
Time differences

Presentation
Time calculations

Worksheets
Time
Time calculations
World time zones

Puzzles
Time calculations
Australian times

Video
Time differences

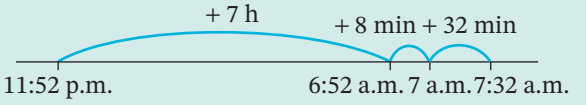
Example 21

What is the time 7 hours and 40 minutes after 11:52 p.m.?

Solution

7 hours after 11:52 p.m. is 6:52 a.m.

40 minutes after 6:52 a.m. is 7:32 a.m.



Example 22

a Ms Hossain bought a new car in August 2016. How old was her car in March 2022?

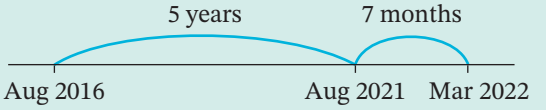
b Boris was born on 14 April 2007. How old is he on 29th September 2025?

Solution

a From August 2016 to August 2021 = 5 years

From August 2021 to March 2022 = 7 months

Age of car = 5 years 7 months

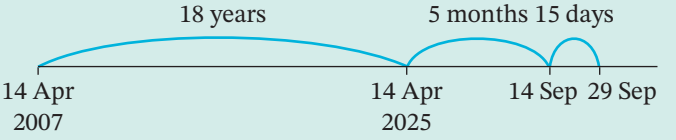


b From 14/4/07 to 14/4/25 = 18 years

From 14/4/25 to 14/9/25 = 5 months

From 14/9/25 to 29/9/25 = 15 days

Boris' age = 18 years 5 months 15 days



EXERCISE 12.10 ANSWERS ON P. 606

Time differences U F PS R C

EXAMPLE 19

1 What is the difference in time between 10:42 a.m. and 2:13 p.m.?
Select the correct answer **A**, **B**, **C** or **D**.

A 3 h 31 min
B 4 h 55 min
C 8 h 29 min
D 12 h 55 min

2 Calculate the time difference between:

a 7:15 p.m. and 8:20 p.m.
b 10:16 a.m. and 12:06 p.m.
c 4:09 a.m. and 9:53 a.m.
d 11:15 p.m. and 3:08 a.m.

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e 7:27 a.m. and 1:12 p.m.
f 9:36 p.m. and 9:14 a.m.
g 7:45 p.m. and 10:10 p.m.
h 2:24 a.m. and 3:07 a.m.
i 4:15 p.m. and 6:02 p.m.
j 10:25 a.m. and 2:33 p.m.
k 8:40 a.m. and 4:19 p.m.
l 6:45 a.m. and 8:10 p.m.

3 Calculate the time difference between: EXAMPLE 20

a 08:00 and 11:00
b 05:00 and 15:00
c 04:50 and 19:30
d 06:30 and 10:45
e 09:40 and 14:55
f 11:20 and 20:05
g 05:10 and 12:15
h 07:40 and 13:10
i 12:45 and 21:25
j 13:40 and 21:50
k 10:35 and 16:30
l 09:45 and 18:25

4 What time will it be: EXAMPLE 21

a 5 hours after 3:00 p.m.?
b 8 hours after 11:00 a.m.?
c 28 minutes after 7:15 p.m.?
d 3 hours 32 minutes after 9:45 a.m.?
e 3 hours 19 minutes after 10:49 p.m.?
f 4 hours after 9:32 a.m.?
g 9 hours after 5:14 p.m.?
h 45 minutes after 3:30 p.m.?
i $2\frac{1}{4}$ hours after 4:02 a.m.?
j 12 hours 40 minutes after 2:45 a.m.?

5 A marathon began at 10:20 a.m. Here are 6 of the competitors and the times they ran. Write the runners in their order of finishing and the time each crossed the finishing line. R C

Mike 3:11 (3 h 11 min)
Joe 2:23
Anna 2:54
Pathena 3:01
Ken 2:59
Gail 3:42

6 A plane left Perth at 09:00 and arrived in Sydney at 13:45 (Perth time). How long was the journey?

7 At Westvale Islamic College, the school day begins at 8:45 a.m. and ends at 3:10 p.m. How long is the school day?

8 What time will it be:

a 4 hours before 6:15 p.m.?
b 7 hours before 9:30 a.m.?
c 45 minutes before 3:20 p.m.?
d 2 hours 10 minutes before 1:35 p.m.?
e 3 hours 30 minutes before 11:25 a.m.?
f 1 hour 25 minutes before 7:40 p.m.?

9 Jamie bought a computer in June 2019. How old is it in: R EXAMPLE 22

a August 2023?
b January 2024?

10 Darika was born on 15th March 2013. How old is she in years, months, days on: R

a 20th April 2019?
b 9th September 2028?

11 A movie starts at 1:40 p.m. and ends at 3:55 p.m. What is the length of the movie?

12 A box of chocolates bought in November 2024 has a 'Use by' date of March 2026. How long will it last?

13 Mr Badger started teaching at Westvale High School in February 1998. How long had he taught at the school for in December 2025?

14 Karen works at the office from 8:10 a.m. till 4:30 p.m. How much will she earn for the day if her hourly wage is $28.11? PS

15 South Australia is $1\frac{1}{2}$ hours ahead of Western Australia. Anne is flying from Perth to Port Augusta in South Australia. If the flight takes $2\frac{1}{2}$ hours and it leaves Perth at 10 a.m., what will the local time be in Port Augusta when the plane lands? PS R

16 Will and Kate were married on 29th April 2011. How long will they have been married in years, months and days on 10th May 2025? R

12.11 Timetables

Worksheet
Tide chart

Example 23

This part of a timetable is a schedule for a bus travelling between Sydney and Wagga Wagga.

Sydney to Wagga Wagga		Wagga Wagga to Sydney	
Sydney	2:30 p.m.	Wagga Wagga	7:15 a.m.
Strathfiel	3:00 p.m.	Gundagai	8:25 a.m.
Yagoona	3:20 p.m.	Jugiong	8:54 a.m.
Liverpool	3:45 p.m.	Yass	9:41 a.m.
Mittagong	4:40 p.m.	Goulburn*	10:41 a.m.
Goulburn*	5:40 p.m.	Mittagong	12:10 p.m.
Yass	7:10 p.m.	Liverpool	1:05 p.m.
Jugiong	7:55 p.m.	Yagoona	1:20 p.m.
Gundagai	8:20 p.m.	Strathfiel	1:35 p.m.
Wagga Wagga	9:30 p.m.	Sydney	2:05 p.m.

* 30 minute meal stop at Goulburn

a At what time does the bus from Sydney arrive in Goulburn?

b Where is this bus at 8:20 p.m.?

c How long does it take to travel from Liverpool to Mittagong?

d Tim is waiting at Yass at 9:15 a.m. for the bus to Sydney. How long will he have to wait before the bus arrives?

Solution

a The bus from Sydney arrives in Goulburn at 5:40 p.m.

b At 8:20 p.m., the bus arrives in Gundagai.

c From Liverpool at 3:45 p.m. to Mittagong at 4:40 p.m., the journey takes 55 minutes.

d Tim needs to wait from 9:15 a.m. to 9:41 a.m., a total of 26 minutes.

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EXERCISE 12.11 ANSWERS ON P. 607

Timetables U F PS R C

EXAMPLE 23

12.11

1 Refer to the bus timetable from the previous example. PS R

a How long does the trip from Sydney to Wagga Wagga take?

b How long would the trip take without a meal break?

c At what time does the bus from Sydney arrive in Yagoona?

d Ali joins the return bus at Jugiong and gets off at Liverpool. How long is his trip?

e How long does Cherie have to wait for the bus to Sydney if she arrives in Strathfield at 1:22 p.m.?

f Find the time taken from Liverpool to Sydney and from Sydney to Liverpool. Suggest a reason for the difference.

2 Chris' volleyball team flies from Sydney to Brisbane for a tournament. Chris visited the NelsonAir website and found the following daily flight schedule. PS R C

Flight no.	Sydney	Brisbane
TH503	09:05	10:30
TH511	09:35	11:00
TH038	10:05	11:30
TH114	10:40	12:10
TH514	11:05	12:30
TH051	11:35	13:00

a When does flight TH511 leave Sydney and how long is the journey?

b What and when is the latest flight before 11 a.m.?

c The team plans to meet at Sydney airport at 9:40 a.m. How long will they need to wait for the next available flight?

d The team needs to be at Sydney airport 40 minutes before a flight takes off. When should they be at the airport to catch the TH038 flight?

e The team needs to be at their hotel in Brisbane by 12:30 p.m. If it takes 30 minutes to drive from the airport to the hotel, what is the latest flight the team can catch from Sydney?

f Which flight takes longer to reach Brisbane than the others? Give one reason why it might take longer.

3 This section of a train timetable shows the Goulburn to Sydney run. PS R C

Goulburn to Sydney — Monday to Friday								
	am	am	am	pm	pm	pm	pm	pm
GOULBURN	5:08	7:27	8:17	1:47	2:45	4:26	6:47	7:45
MARULAN	5:26	7:45			3:03			8:03
TALLONG	5:32	7:51	Bookings	Bookings	3:09	Bookings	Bookings	8:09
WINGELLO	5:39	7:58	essential	essential	3:16	essential	essential	8:16
PENROSE	5:44	8:03			3:21			8:21
BUNDANOON	5:50	8:09	8:52	2:22	3:27		7:22	8:27
EXETER	5:55	8:14			3:32			8:32
MOSS VALE	6:05	8:24	9:05	2:35	3:42	5:13	7:35	8:42
BURRADOO	6:10				3:47			8:47
BOWRAL	6:13	8:30	9:11	2:41	3:50		7:41	8:50
MITTAGONG	6:17	8:34	9:16	2:46	3:54		7:46	8:54
YERRINBOOL	6:30				4:07			9:07
BARGO	6:41				4:18			9:18
TAHMOOR	6:48				4:25			9:25
PICTON	6:56	9:08			4:33			9:33
CAMPBELLTOWN	7:23	9:30	10:11	3:41	5:00	6:13	8:42	10:00
STRATHFIELD			10:42	4:17		7:00	9:12	
SYDNEY	8:12	10:12	10:54	4:29	6:20	7:13	9:24	11:04

a Georgina travels from Penrose to Yerrinbool, arriving at 4:07 p.m. How long did the trip take?

b Michael has an interview in Sydney on Tuesday at 10:45 a.m. At what time must he catch the train in Goulburn?

c What is the difference in the time taken to travel from Goulburn to Sydney on the 5:08 a.m. train and the 8:17 a.m. train?

d You have been visiting friends in Moss Vale and are returning to Sydney. Decide which train you would catch and explain why.

e A new train is added to the timetable, leaving Goulburn at 11:12 a.m. Write out a timetable for this train if it stops at the same stations as the 6:47 p.m. train.

4 An Explorer Bus is a shuttle bus that takes tourists on a tour of a city and allows them to visit places of interest. PS R C

Explorer depot	10:00	10:25	10:50	11:15	11:45	12:00	12:25	12:50	1:15
City cathedral	10:08	10:33	10:58	11:23	11:53	12:08	12:33	12:58	1:23
Railway station	10:15	10:40	11:05	11:30	12:00	12:15	12:40	1:05	1:30
Parliament	10:24	10:49	11:14	11:39	12:09	12:24	12:49	1:14	1:39
Museum	10:35	11:00	11:25	11:50	12:20	12:35	1:00	1:25	1:50
City square	10:45	11:10	11:35	12:00	12:30	12:45	1:10	1:35	2:00
Zoo	11:00	11:25	11:50	12:15	12:45	1:00	1:25	1:50	2:15
Town Hall	11:12	11:37	12:02	12:27	12:57	1:12	1:37	2:02	2:27
Arts centre	11:19	11:44	12:09	12:34	1:04	1:19	1:44	2:09	2:34
Water gardens	11:30	11:55	12:20	12:45	1:15	1:30	1:55	2:20	2:45
Hall of fame	11:38	12:03	12:28	12:53	1:23	1:38	2:03	2:28	2:53
Explorer depot	11:50	12:15	12:40	1:05	1:35	1:50	2:15	2:40	3:05

a The bus leaves the railway station at 11:30 a.m. When will it arrive at the Arts centre and how long will the journey take?

b Harrison needs to be at the Hall of fame by 2:30 p.m. What is the latest time he needs to get to Parliament to achieve this?

c Vo, Binh and Vicki came to the city by train, arriving at the station at 11:42 a.m. They caught the Explorer Bus to the zoo. What is the earliest time they could expect to arrive at the zoo?

d What is the smallest number of buses needed to meet this Explorer Bus timetable? Explain how you arrived at your answer.

e Manuel and Sofia arrived at the City cathedral at 10:25 a.m. They arrange to meet their hosts at the Hall of fame at 2:45 p.m. They want to spend at least half an hour at the museum, photograph the City square and have lunch at the Town Hall. Plan a list of times for them to catch the Explorer Bus to do these things and meet their hosts on time.

f In summer, extra Explorer tours leave the depot at 11:30 a.m. and 1:30 p.m. Make a list of departure times that appear in the timetable for the 1:30 p.m. tour.

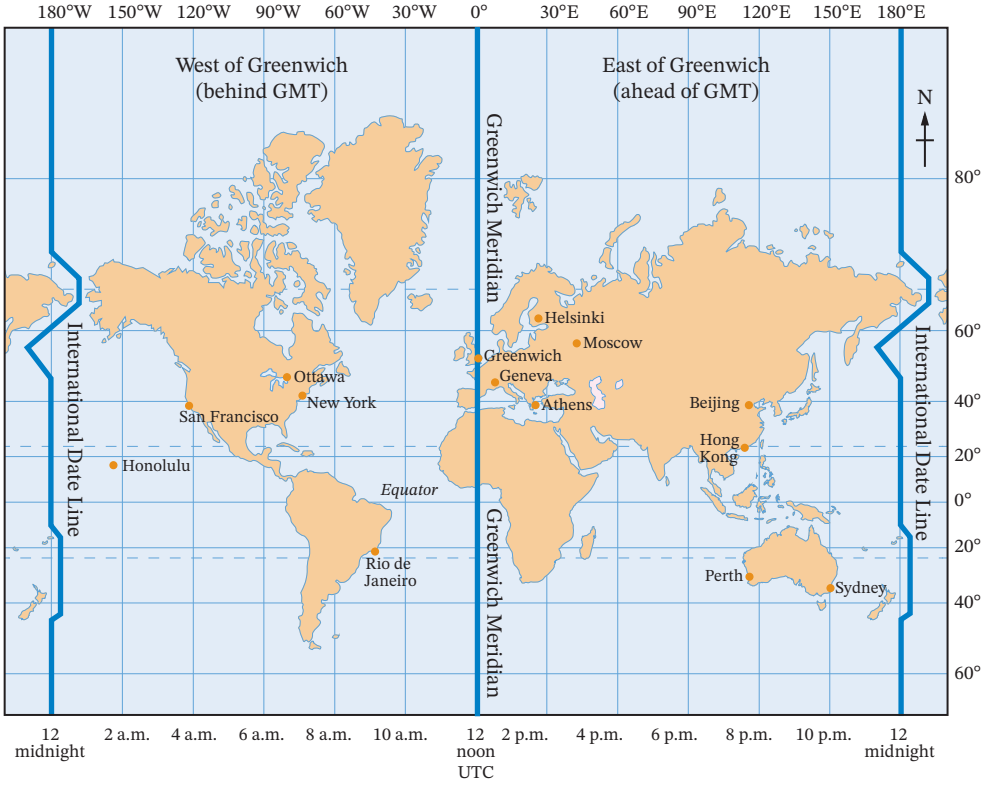
12.12 Extension: World time zones

EXTENSION

Worksheet
World time zones

Puzzle
Australian times

The world is divided into 24 hourly time zones. Time is the same throughout each zone. The centre of each time zone is a meridian of longitude (an imaginary line running from the North Pole to the South Pole). Each hourly time zone covers 15° of longitude.



The map above shows how times around the world are related. All time is measured in relation to the time at the Greenwich Observatory (near London), either ahead of or behind **UTC (Coordinated Universal Time)**, also known as **Greenwich Mean Time (GMT)**. Greenwich is pronounced 'Grennitch'. Time zones in Australia are ahead of UTC because Australia is east of Greenwich. Time zones in the USA are behind UTC because the USA is west of Greenwich.

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EXERCISE 12.12 ANSWERS ON P. 607

EXTENSION

World time zones U F PS R C

1 Use the map above to determine whether each city is ahead of or behind UTC. R

a	Sydney	**b**	San Francisco	**c**	Rio de Janeiro	**d**	Perth
e	Beijing	**f**	Honolulu	**g**	Moscow	**h**	Athens
i	Hong Kong	**j**	Helsinki	**k**	New York	**l**	Ottawa

2 Find the time in each city when it is 12 midday in Greenwich.

a	Sydney	**b**	Perth	**c**	New York	**d**	Beijing
e	San Francisco	**f**	Honolulu	**g**	Moscow	**h**	Geneva

3 What is the time difference between:

a	Sydney and Perth?	**b**	Sydney and Beijing?
c	Sydney and Honolulu?	**d**	Sydney and Moscow?
e	Sydney and New York?	**f**	Perth and Beijing?
g	San Francisco and New York?	**h**	Honolulu and Moscow?
i	Geneva and Perth?	**j**	San Francisco and Geneva?

4 If it is 2:00 p.m. in Sydney, what is the time in:

a	Greenwich?	**b**	Perth?	**c**	New York?	**d**	Beijing?
e	San Francisco?	**f**	Honolulu?	**g**	Moscow?	**h**	Geneva?

5 A cricket match being played in India is telecast live at 7:00 p.m. Sydney time. What is the local time of the cricket match if Sydney's time is $4\frac{1}{2}$ hours ahead of India's? PS R

6 Simone, in Newcastle (NSW), wants video call her cousin Zac in Vancouver, Canada. The time in Vancouver is 18 hours behind the time in Newcastle. At what time should Simone call to catch Zac when it is 3:00 p.m. in Vancouver? PS R

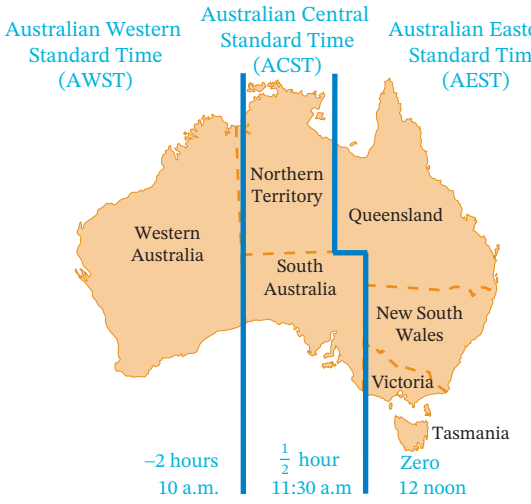
7 Brisbane is 2 hours behind Auckland. A plane leaves Auckland at midday and takes 3 hours to fly to Brisbane. What is the local time in Brisbane when the plane lands? Select the correct answer **A**, **B**, **C** or **D**. PS R

A	11 a.m.	**B**	1 p.m.	**C**	3 p.m.	**D**	5 p.m.

8 Find out what happens if you cross the International Date Line (IDL). Why isn't the IDL straight? R C

EXTENSION

9 This map shows the 3 time zones for Australia.



State whether each city is ahead of, behind or has the same time as Adelaide (SA).

a Sydney **b** Melbourne **c** Darwin
d Perth **e** Mt Isa (Qld) **f** Geraldton (WA)
g Cobar (NSW) **h** Ceduna (SA) **i** Cairns (Qld)

10 What is the time difference between:

a Sydney and Adelaide? **b** Melbourne and Perth?
c Adelaide and Melbourne? **d** Hobart and Darwin?
e Canberra and Perth? **f** Brisbane and Canberra?

11 If it is 11 p.m. in Sydney, what time is it in:

a Melbourne? **b** Adelaide? **c** Perth?
d Darwin? **e** Hobart? **f** Canberra?

12 If it is 5:30 p.m. in Adelaide, what time is it in:

a Melbourne? **b** Sydney? **c** Perth?
d Darwin? **e** Hobart? **f** Brisbane?

13 a Jason flies from Sydney to Perth, taking 4 hours. If he leaves Sydney at 2 p.m., at what time does he land in Perth? Give your answer as Perth local time.

b When Jason flies home, he leaves Perth at 9 a.m. At what time does he land in Sydney? Give your answer as Sydney local time. PS R

14 a Find out when daylight saving begins and ends.

b Why do we have daylight saving?

c How does daylight saving affect the different time zones?

d If it is 12:30 p.m. in Western Australia (not on daylight saving), what time is it in New South Wales on Eastern Standard Daylight Saving Time? PS R C

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EXTENSION

Investigation

World trip

Plan a trip around the world with at least 3 stopovers (for example, Tokyo, Hanoi, Cairo). Use airline timetables so you can give details of departures, arrivals and the length of each flight. Does it matter if you head east or west when you start? What effect does the International Date Line have on your trip?

Did you know?

World time zones

During the 19th century, when there was little world travel, cities and countries operated on local time. However, as shipping and business increased between different places around the world, the need for a standard time measurement arose. World time zones were first proposed by a Scottish Canadian engineer, Sir Sandford Fleming, in 1878.

Hourly time zones are based on longitude. 15° difference in longitude across the world equals 1 hour difference in local time. Time is measured in relation to the Greenwich Observatory near London. At the time, Britain was the leading country in shipping and the observatory produced high quality data. Sydney is 10 hours ahead of Greenwich, New York is 5 hours behind. This makes the time difference between Sydney and New York 15 hours.

How is time measured at the North and South poles?



Shutterstock.com/Lefteris Papaulakis

The Greenwich Observatory

Power plus ANSWERS ON P. 607

1 A spider moves at 1 cm/s. If it starts at the back left-hand corner of your classroom, find how long it will take to reach:

a the nearest person **b** you

c the whiteboard **d** the door

2 Simplify each ratio:

a 0.4 : 0.7 **b** 1.3 : 0.8 **c** 0.5 : 0.05

d 0.49 : 1.4 **e** $\frac{1}{3} : \frac{1}{8}$ **f** $\frac{3}{5} : \frac{1}{2}$

3 Kobe and Hashim renovated their gardens. Kobe planted 19 native plants and 14 non-native plants. Hashim put in 24 native plants and 20 non-native plants. Kobe believes that her garden is more Australian native than Hashim's. Is Kobe right? Explain your answer.



Shutterstock.com/Dorothy Chiron

4 Australia's annual birth rate in 2023 was approximately 12.1 per 1000. Given that the population then was about 26 717 000, how many babies were born in 2023?

5 Singapore is a small island but it has a population of 6 106 700, making it the 2nd-most densely populated country in the world. Its population density is 8381 persons/km^2.

a What is the area of Singapore, to the nearest square kilometre?

b Australia is the 4th least densely populated country in the world, at a rate of only 3.4 persons/km^2 and an area of 7 682 300 km^2. If Australia was as densely populated as Singapore, what would its population be, to the nearest hundred?

6 Mae invents a decimal system of time, where 1 minute = 100 seconds and 1 hour = 100 minutes. One second is the same as our second, but decimal minutes and hours are different to ours.

a How many of our minutes will there be in a decimal hour?

b How many decimal hours will there be in one of our days?

CHAPTER 12 REVIEW

Language of maths

12-hour time	24-hour time	best buy	equivalent ratios
per	rate	ratio	simplify
speed	steepness	term	timetable
travel graph	unit price	unitary method	

1 What is **unit price** and what is it used for?

2 Which feature of a travel graph indicates the speed of the traveller?

3 What is the difference between a **ratio** and a **rate**?

4 How do you simplify:

a a ratio? **b** a rate?

5 Why does the 'unitary method' have that name?

6 What are the advantages of using **24-hour time**?

7 How do you say the '/' symbol in a rate?

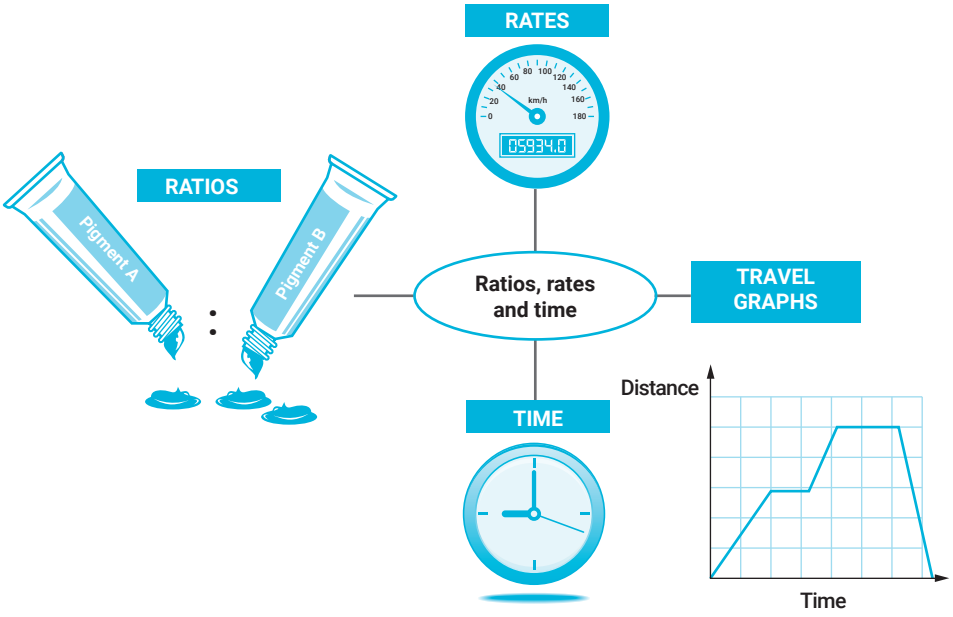
Quiz
Language of maths 12

Puzzle
Ratios and rates find-a-word

Topic summary

- Give examples of places in which ratios, rates or time are used.
- Give some examples of situations where you would use what you know about ratio, rates and time.
- What parts of this chapter were new to you?
- What did you find most difficult about this topic? Discuss any problems with your teacher or with a friend.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



Worksheet
Mind map: Ratios, rates and time

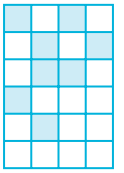
Quiz
Test yourself 12

TEST YOURSELF 12

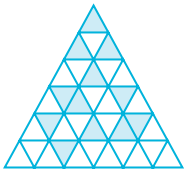
ANSWERS ON P. 607

12.01 **1** In each diagram, find the ratio of shaded parts to unshaded parts.

a



b



12.01 **2** In a construction business, there are 17 women and 38 men. Find the ratio of:

a men to women **b** women to the total

12.02 **3** Copy and complete each pair of equivalent ratios.

a 2 : 3 = 8 : ___ **b** 1 : 5 = 2 : ___ **c** 2 : 3 : 4 = 6 : ___ : ___

12.02 **4** Simplify each ratio.

a 12 : 21 **b** 25 : 75 **c** 6 : 36

d 25 : 45 **e** 18 : 6 : 24 **f** 5 : 25 : 100

g 2 cm : 10 mm **h** 5 dozen eggs : 6 eggs **i** 3 L : 600 mL

j 50 m : 1 km **k** 20 minutes : 1 hour **l** 40 cents : $2

m 18 months : 3 years **n** 12 days : 3 weeks **o** 4 days : 10 hours

12.03 **5** **a** The ratio of girls to boys in Year 7 is 4 : 3. If there are 75 boys in Year 7, find how many girls there are.

b Tyrone and Erin invested in a business in the ratio 5 : 7. If Erin invested $63 000, how much did Tyrone invest?

12.04 **6** Write each statement as a rate.

a 150 000 voters for 50 politicians **b** 1000 sheep over 5 hectares

c 10 tickets for $8 **d** 810 stitches to make 3 balls

12.05 **7** For each pair, determine which is the better buy.

a Dog biscuits: 400 g of Woof for $5.20 or 1 kg of Chow for $11.10.

b Milk: 1 L of Dairy Farm for $2.85 or 1250 mL of Milky for $3.90.

12.06 **8** **a** Marie earns $22.45 per hour. How much is she paid for 38 hours work?

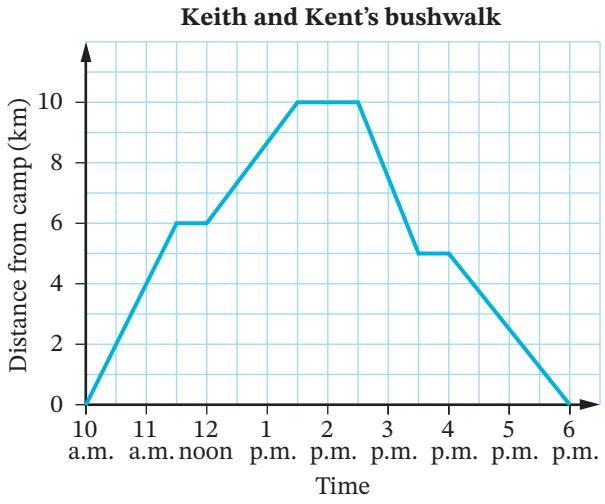
b How many litres of petrol can you buy with $65 if petrol costs $1.79/L? Answer correct to one decimal place.

c Fertiliser is used at 20 kg/ha. How many hectares can be covered with 150 kg of fertiliser?

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9 Keith and Kent decided to go bushwalking one day. This travel graph shows their journey. 12.07



a How far did Keith and Kent walk?

b How many stops did they make?

c At what time did they start back?

d Between what times were Keith and Kent walking the fastest?

e Calculate their speed in the last 2 hours.

10 Round each time to the nearest hour. 12.08

a 9 h 50 min **b** 3.2 h **c** 4 h 12 min 49 s

11 Convert: 12.08

a 316 minutes to hours and minutes

b 208 seconds to minutes and seconds

12 Simplify each expression. 12.08

a 6 h 45 min + 3 h 20 min **b** 3 h 16 min + 1 h 26 min

c 4 h 33 min – 2 h 24 min **d** 2 h 19 min – 1 h 50 min

13 Convert to 12-hour time. 12.09

a 01:48 **b** 06:24 **c** 15:50 **d** 12:30

14 Convert to 24-hour time. 12.09

a 2:10 p.m. **b** 5:33 a.m. **c** 12:19 a.m. **d** 1:50 p.m.

15 Calculate the time difference between each pair of times. 12.10

a 5:25 a.m. and 9:45 a.m. **b** 11:55 p.m. and 7:30 a.m.

c 1:15 p.m. and 8:10 p.m. **d** 07:50 and 14:20

e 23:45 and 00:15 **f** 15:30 and 03:25

16 Samantha was born on 3rd June 2004. Calculate her age in years, months and days on 16th August 2023. 12.10

17 Refer to the train timetable on page 536. 12.11

a How long does the 7:45 p.m. train take to get to Sydney?

b When is the latest time you need to catch the train at Moss Vale to be in Sydney by 11 a.m.?

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PRACTICE SET 4

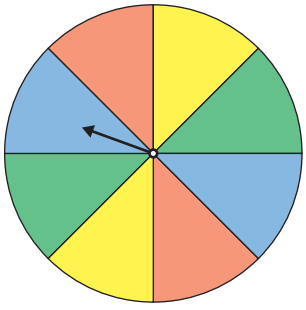
ANSWERS ON P. 607

11.01

1 List the sample space for each experiment.

a Rolling a die

b Spinning this spinner



12.01

2 There are 18 girls in a class of 30 students.
Write in simplest form the ratio of:

a girls to students

b girls to boys

c students to boys

12.04

3 Simplify each rate.

a 200 km in 8 hours

b $24 for 6 metres

c 1000 revolutions in 2 minutes

d 2100 cars in 7 days

12.10

4 How many hours and minutes between:

a 4:30 a.m. to 9:55 a.m.?

b 7:15 p.m. to 8:25 p.m.?

c 9:24 p.m. to 9:05 a.m.?

d 15:05 to 22:10?

12.09

5 Write in 24-hour time:

a 8:25 a.m.

b 4:45 p.m.

12.09

6 Write in 12-hour time:

a 21:15

b 04:30

10.03

7 **a** Construct a dot plot for these marks for a test out of 25:

11	8	12	12	15	13	10	12	13	15	16
22	12	11	11	10	13	16	12	15	12	11

b Where are the marks clustered?

c Identify any outliers.

10.07

8 Find the range of the marks in question **7**. Select the correct answer **A**, **B**, **C** or **D**.

A 8 **B** 12 **C** 14 **D** 22

12.02

9 Simplify each ratio.

a 2 : 6 **b** 96 : 36 **c** 24 : 72

d 2 m to 15 cm **e** 10 minutes to 2 hours **f** 12 : 30 : 42

12.06

10 In 20 hours, a train travels 800 km.

a What is its average speed in km/h?

b How far would it travel in 11 hours at the same rate?

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11 A die is rolled. Find the probability of rolling: 11.02

a an even number **b** a number greater than 2.

12 Copy and complete each conversion. 12.08

a 3 fortnights = _____ days **b** 96 hours = _____ days

c $2\frac{1}{2}$ years = _____ weeks **d** 105 days = _____ weeks

e 212 minutes = _____ hours _____ minutes

f 330 seconds = _____ minutes _____ seconds

13 At a vehicle checkpoint, the following types of vehicles were counted in a 2-hour period. 11.04

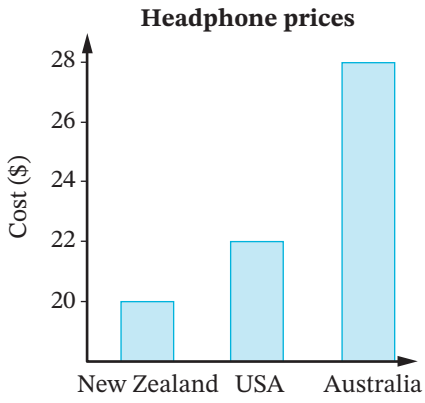
Cars 30 Trucks 7 Motorcycles 11 Buses 2

a How many vehicles passed the checkpoint in this time period?

b Based on this data, what is the probability that the next vehicle to pass the checkpoint will be a car?

14 **a** Explain why this graph gives a misleading impression. 10.02

b Redraw the graph correctly.



15 A packet of mixed lollies has 12 red lollies, 9 green lollies, 4 black lollies and 5 pink lollies. You tip the packet and one lolly rolls out. What is the probability that it is not a green lolly? Select **A**, **B**, **C** or **D**. 11.02

A $\frac{4}{30}$ **B** $\frac{5}{30}$ **C** $\frac{12}{30}$ **D** $\frac{21}{30}$

16 Use the timetable on page 536 to answer this question. Louella catches the train from Bundanoon to Sydney to meet up with friends. Which train gives her the shortest trip? Select **A**, **B**, **C** or **D**. 12.11

A 8:09 a.m. **B** 8:52 a.m. **C** 2:22 p.m. **D** 3:27 p.m.

10.04

17 Organise this set of data into an ordered stem-and-leaf plot:

9	5	15	22	11	14	19	28	18
13	17	6	24	16	15	4	10	21

10.07

18 Find the mode of the data in question **17**. Select **A**, **B**, **C** or **D**.

A 5 **B** 15 **C** 18 **D** 24

12.03

19 The ratio of boys to girls at the public speaking competition is 4 : 3.
If there are 15 girls, find the number of boys.

11.03

20 A pack of 20 cards contains 10 blue, 6 purple and 4 pink cards. One card is drawn from the pack at random. Find the probability that this card is:

a purple **b** not purple

10.05

21 For this set of scores:

3.3 3.5 3.0 3.0 3.8 3.4 3.5 3.9

10.06

find:

a the mean **b** the mode

c the median **d** the range

12.06

22 Water flows into a tank at the rate of 15 litres per hour. How long will it take to fill the tank if it holds 750 litres?

12.05

23 Yummo ice cream comes in 3 sizes:

- 1 litre for $3.25
- 2 litres for $6.30
- 1.5 litres for $4.80.

Which size is the best buy?

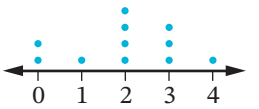
10.07

24 For each set of data, find:

i the range **ii** the mode

iii the median **iv** the mean (to 2 decimal places).

a Number of goals scored



b Ages of soccer players

Stem	Leaf
1	5 7 9 9
2	1 1 1 2
3	0

11.05

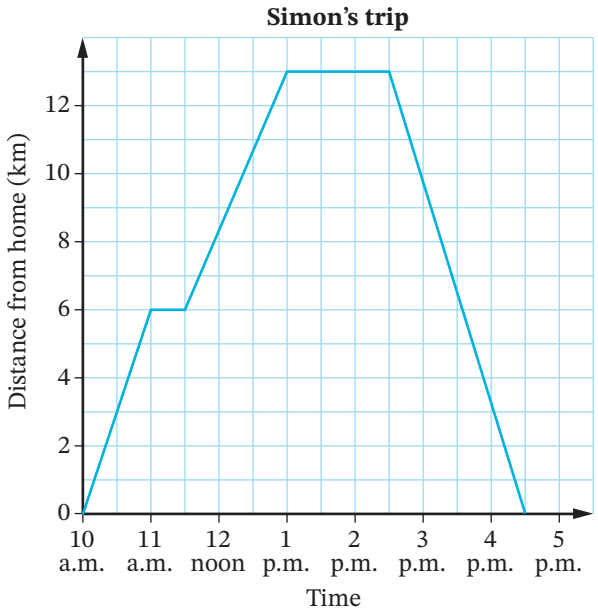
25 The letters of the word MATHEMATICS are written on separate cards and one is randomly selected. What is the probability that the letter drawn out is:

a not S? **b** not a vowel? **c** neither M nor H?

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26 Simon's cycling trip is shown on this travel graph. 12.07



a How far was Simon from home at 11 a.m.?

b When, and for how long, did Simon stop for lunch?

c How far did Simon cycle in total?

d At what times was Simon 3 km from home?

e Was Simon's speed greater in the first hour or between 11:30 a.m. and 1 p.m.?

27 Manal's dad records the number of phone calls she makes each night for one week: 10.05

9 3 6 2 13 # 1

where # represents a missing value.

What is # if the mean of the values is 5? Select **A**, **B**, **C** or **D**.

A 1 **B** 2 **C** 5 **D** 6

28 Jane and Neil were arguing about which local rugby league team is the best. The points scored by their favourite teams in each match over the previous season were: 10.08

Eagles: 20, 10, 40, 12, 17, 20, 22, 20, 34, 19, 36, 18, 24, 12, 38, 34, 24, 36, 32, 22, 6, 7, 38, 18

Cougars: 14, 18, 24, 39, 14, 4, 4, 14, 10, 13, 28, 22, 16, 18, 18, 12, 18, 28, 21, 6, 10, 18, 36, 12

a Draw a back-to-back stem-and-leaf plot for these 2 data sets.

b What is the highest number of points scored by either team?

c What is the lowest number of points scored by either team?

d Find the mode for each team.

e Which is the better team based on the data? Justify your answer.

f What other data might you need to know to decide which is the better team?